STAT 210A: Theoretical Statistics

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Contents

8/27/26 Lecture 1: Introduction . . . 1
9/2/26 Lecture 2: What is Probability? . . . 2
Definition 2.1: σ-fields . . . 2
Definition 2.2: Measures . . . 3
9/3/26 Lecture 3: Probability Spaces and Statistics . . . 5
Definition 3.1: Absolute Continuity with respect to a Measure . . . 5
Theorem 3.2: Radon-Nikodym Theorem . . . 5
Definition 3.3: Almost Everywhere Equivalence . . . 5
Definition 3.4: Random Variables . . . 6
Definition 3.5: Statistical Models . . . 6
Definition 3.6: Parametric Models . . . 6
Definition 3.7: Non-parametric Models . . . 7
9/8/26 Lecture 4: Estimation and Sufficiency . . . 8
Definition 4.1: Inadmissable Estimators . . . 8
Definition 4.2: Sufficient Statistics . . . 9
Theorem 4.3: Factorization Theorem . . . 10
9/10/26 Lecture 5: More on Sufficiency . . . 12
Definition 5.1: Order Statistics . . . 12
Definition 5.2: The Dirac Measure . . . 13
Definition 5.4: Minimal Sufficient Statistic . . . 13
Theorem 5.5 (Bahader) . . . 14
Definition 5.6: Likelihood Functions . . . 14
9/15/26 Lecture 6: Exponential Families . . . 15
Definition 6.1: Exponential Families . . . 15
Definition 6.2: Natural Parameter Space . . . 15
9/17/26 Lecture 7: Exponential Families . . . 19
Definition 7.2: Canonical Form . . . 21
Definition 7.3: Minimal Form . . . 21
Definition 7.5: Completeness of a Sufficient Statistic . . . 22
9/22/26 Lecture 8: Complete Statistics . . . 24
Definition 8.1: Completeness . . . 24
Definition 8.2: Full-Rank and Curved Exponential Families . . . 25
Theorem 8.3: Full Rank Families have Complete Sufficient Statistics . . . 25
Theorem 8.4: Complete Sufficient implies Minimal Sufficient . . . 25
Defintion 8.5: Ancillary Statistics . . . 26
Theorem 8.6: Basu's Theorem . . . 26
9/26/26 Lecture 9: Rao-Blackwellization . . . 28
Definition 9.1: Convex Functions . . . 28
Theorem 9.2: Jensen's Inequality . . . 28
Theorem 9.3: Rao-Blackwell Theorem . . . 29
Definition 9.4: U-estimability . . . 29
Definition 9.5: Uniform Minimum Variance Unbiased (UMVU) . . . 29

Theorem 9.6: Uniqueness of Rao-Blackwellization . . . 29
9/29/26 Lecture 10: Score and Fisher Information . . . 33
Definition 10.1: Score . . . 34
Definition 10.2: Fisher Information . . . 34
10/1/26 Lecture 11: Cramer-Rao Lower Bound . . . 36
Definition 11.1: Bayes Risk . . . 38
Definition 11.2: Posterior . . . 38
10/6/26 Lecture 12: More on Bayes Estimators . . . 39
10/8/26 Lecture 13: Conjugate Priors . . . 42
Definition 13.1: Conjugate Priors . . . 43
10/13/26 Lecture 14: Objective Bayes . . . 45
Definition 14.1: Flat Prior . . . 45
Definition 14.2: Jeffreys' Prior . . . 45
10/15/26 Lecture 15: Hierarchical Models and Markov Chain Monte Carlo . . . 48
Definition 15.1: Stationary Markov Chains . . . 50
Definition 15.2: Stationary Distributions . . . 50
Definition 15.3: Detailed Balance . . . 50
Theorem 15.5 Markov Chain Convergence Theorem . . . 51
10/20/26 Lecture 16: Gibbs Sampling in Practice and Empirical Bayes . . . 52
10/22/26 Lecture 17: The James-Stein Estimator and Stein's Lemma . . . 55
Theorem 17.2: Stein's Lemma, Univariate . . . 56
Definition 17.3: Jacobians . . . 57
Definition 17.4: Frobenius Norm . . . 57
Theorem 17.5: Stein's Lemma, Multivariate . . . 57
10/27/26 Lecture 18: Beyond James-Stein and Minimax Estimation . . . 59
Definition 18.1: Minimax Risk . . . 59
Definition 18.2: Least Favorable Priors . . . 60
Theorem 18.3: Correspondence between Bayes and Minimax . . . 60
10/29/26 Lecture 19: Least Favorable Sequences and Hypothesis Testing . . . 63
Definition 19.1: Least Favorable Sequences . . . 63
Theorem 19.2 . . . 63
Definition 19.3: Simple and Composite Hypotheses . . . 64
Definition 19.4: Level-α Tests . . . 66

8/27/26 Lecture 1: Introduction

Deductive Reasoning: Drawing conclusions from premises ("axioms"). This is non-ampliative, meaning no new information is created. Using deductive reasoning is "risk-free" in the sense that if the premises are true, all conclusions are true as well.

Inductive Reasoning: Using observations to make general claims. This is ampliative, meaning that you are creating new information. However, there are "risks" to inductive reasoning, in the sense that one's conclusions may still be wrong even if the premises hold.

- **Problem of Induction (Hume):** How can one justify making an inductive claim? One can argue using the uniformity principle that what happens today will happen tomorrow, but this is justifying inductive reasoning using inductive reasoning! Thus, let's just assume it works.

Now, all of statistics and science is build on inductive reasoning, so how do we even make sure the claims we make are valid (in some sense)? The way to do this is by turning it into a deductive reasoning problem, and there are two main ways to do that.

- **Idea #1 Bayesian Reasoning:** Start with prior beliefs and use mathematics and probability theory to update these beliefs.
  - **Pros:** We can actually say something like "there is a 95% chance our hypothesis is true"
  - **Cons:** We must make an assumption on what the prior beliefs are, which we cannot justify.
- **Idea #2 Inductive Behavior (Frequentist):** Reason directly from the data, and quantize error.
  - **Pros:** No assumptions needed, relies strictly on data.
  - **Cons:** We cannot quantize the probability our hypothesis is true, but instead we can say that a process works %95 of the time under repetition.

9/2/26 Lecture 2: What is Probability?

First, here are the questions that we will be answering in this course:

1. **Bayesian vs Frequentist**: What are the pros and cons? How do different frameworks influence how we view probability?
2. **Sufficiency:** Does a parameter sufficiently summarize the data well? If we were to forget all the data and keep the parameter, is anything of importance lost?
3. **Estimation:** What is a good way to estimate parameters? For a binomial parameters, it seems like X/n is a good estimator, and why? Even further, how can we use outside information to improve our estimators?
4. **Non-Parametric Testing:** How do we test confidence without parameters?
5. **Testing:** Hypothesis testing and rejecting them
6. **Asymptotics:** For large datasets, using certain estimators are not a good idea, due to the sheer computation you would need to do. How do we approximate using asymptotics?

What is Probability?

Frequentist Answer: It is the relative frequency of an outcome when an experiment is repeated many times.

Bayesian Answer: It is the degree of belief that something will happen.

Mathematicians Answer (holy mog): It is a function P that maps some sets of a space X to $[0,1]$ that is countable additive and such that $P(X) = 1$.

Measure Theory Basics

Heuristically, if X is a set, a measure μ is map that maps some subsets of X to the nonnegative reals such that it is countable additive. Here are some examples:

1. Counting Measure: If X is countable, consider the function $\#(A) = \#$ of points in A. This is a measure!

2. Lebesgue Measure: If $X = \mathbb{R}^n$, the *Lebesgue measure* λ is defined by

$$\lambda(A) = \int \cdots \int_A dx_1 \cdots dx_n$$

What does this mean? We will know in a couple weeks maybe.

3. Standard Gaussian: If $X = \mathbb{R}$, let $P_z(A) = \mathbb{P}(z \in A)$ where $z \sim \mathcal{N}(0,1)$. Thus, we see that

$$P_z(A) = \int_A \phi(x)dx \text{ where } \phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

Now, recall we said that measures are only defined over certain collections of sets. We will define what the collection needs to be.

Definition 2.1: σ-fields

A collection $\mathcal{F} \subseteq \mathcal{P}(X)$ is a *σ-field* if it contains X and is closed under complements and countable unions. We then call $(X\mathcal{F})$ a *measurable space*.

For example, if X is countable, then we can have $\mathcal{F} = \mathcal{P}(X)$, and if $X = \mathbb{R}^n$, then we will often have $\mathcal{F} = \mathcal{B}(X)$, where $\mathcal{B}(X)$ is the Borel σ-field, which we will (maybe) define later.

Definition 2.2: Measures

Given a measurable space $(X, \mathcal{F})$, a *measure* μ is a map $\mu : \mathcal{F} \to [0, \infty]$ such that

1. $\mu(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$
2. $\mu(\emptyset) = 0$.

We call μ a *probability measure* if $\mu(X) = 1$. We call $(X, \mathcal{F}, \mu)$ a *measure space*.

Now, if $(X, \mathcal{F}, \mu)$ is a measure space, and $f : X \to Y$ is a nice enough function, we can induce a measure ν on Y as follows: If $B \subseteq Y$, then let

$$\nu(B) = \mu(f^{-1}(B))$$

(thus, we need f to be a nice function in that the preimage of measurable sets are measurable). We call this induced measure the *push-forward measure* on Y.

For instance, if Z is a random variable, and we let $Y = f(Z) = \max Z, 0$, then we see that

$$\begin{aligned} P_Y(B) &= \mathbb{P}(y \in B) \\ &= \mathbb{P}(z \in f^{-1}(B)) \\ &= P_Z(f^{-1}(B)) \end{aligned}$$

Integrals

Heuristically, taking integrals of functions with respect to a measure μ, denote $\int f(x)d\mu(x)$, is a way to put a weight on the area we are integrating over. How do we develop the definition?

First, certainly we want that $\int 1_A(x)d\mu(x) = \mu(A)$ where 1_A is the indicator function of A. Then, because we want linearity, we should allow that

$$\int \sum c_i 1_{A_i}(x)d\mu(x) = \sum c_i \mu(A_i)$$

we call the function $\sum c_i 1_{A_i}(x)$ a simple function. Thus, we currently have the integral defined for indicators and simple functions.

Now, if f is positive, it turns out we can approximate f from below using increasing simple functions, getting more and more accurate. Thus, we can let

$$\int f(x)d\mu(x) = \sup \left\{ \int s d\mu \ : \ s \text{ is simple and } s \leq f \right\}$$

Finally, if f is just any function on $\mathbb{R}$, we can consider it's positive parts $f^+ = \max f, 0$ and $f^- = \max -f, 0$, and as these are integrable in the sense above, define

$$\int f(x)d\mu(x) = \int f^+(x)d\mu(x) - \int f^-(x)d\mu(x)$$

Thus, just from requiring linearity and defining the function on indicators, we have defined the integral for all sorts of functions! Here are some examples:

Example 1: If # is the counting measure, we have that

$$\int f(x)d\#(x) = \sum_{x \in X} f(x)$$

Example 2: If P_Z is the Gaussian distribution, then as

$$\int 1_A(x)dP_Z = \int 1_A(x)\phi(x)dx$$

we see that

$$\int f(x)dP_Z = \int f(x)\phi(x)dx$$

This looks a lot like how expectations are defined in lower level probability classes!

9/3/26 Lecture 3: Probability Spaces and Statistics

Definition 3.1: Absolute Continuity with respect to a Measure

Given a measurable space $(X, \mathcal{F})$ and two measures P, μ, we say that P is *absolutely continuous with respect to* μ if $\mu(A) = 0 \implies P(A) = 0$. We write $P \ll \mu$ and often say that μ *dominates* P.

Recall from before the counting measure, Lebesgue measure, and Standard Gaussian. Then, we see that $P_Z \ll \lambda$, $P_Z \not\ll \#$ as $\#(0.5, 0.75) = 0$ but $P_Z(0.5, 0.75) \neq 0$, and $\# \not\ll \lambda$ as $\lambda\{1\} = 0$ but $\#\{1\} = 1$.

With this definition, we can introduce the following theorem that allows us to define what a density is!

Theorem 3.2: Radon-Nikodym Theorem

If $P \ll \mu$, then we can, under mild conditions, define a *density function* $p : X \to [0, \infty)$ such that

$$\begin{aligned} P(A) = \int_A p(x) d\mu(x) &= \int 1_A(x) p(x) d\mu(x) \\ &= \int 1_A(x) dP \end{aligned}$$

for all measurable A. Thus, we also have that

$$\int f dP = \int f \cdot p d\mu$$

Often, we denote $p(x) := \frac{dP}{d\mu}(x)$ and call it the *Radon-Nikodym derivative.*

Remark: p is not unique, there are many such density functions. For example, with the standard normal, the functions $\phi(x)$ and $\rho(x) = \begin{cases} \phi(x) & x \neq 0 \\ 0 & x = 0 \end{cases}$ are density functions. However, they are unique up to almost everywhere equivalence, which we define next.

Definition 3.3: Almost Everywhere Equivalence

We say f is *equal to* g *almost everywhere* with respect to μ if $\mu(\{x : f(x) \neq g(x)\}) = 0$. We denote this

$$f \overset{\mu \text{ a.e.}}{=} g$$

Now, some common terminology. We say that P is *absolutely continuous* when we mean $P << \lambda$ (λ is the Lebesgue measure). Also, if P is a probability measure, it's density function is called a *probability density function*, and if P is absolutely continuous with respect to the counting measure, then we call the corresponding density function the *probability mass function.*

Probability Spaces and Random Vairables

Let X, Y, Z be random variables. What do we mean when we ask "what is $\mathbb{P}(XY \leq Z^2)$?"

The way we formalize this equation is by assuming these is some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that $X = X(\omega)$, $Y = Y(\omega)$, and $Z = Z(\omega)$. We call $\omega \in \Omega$ an *outcome* and an $A \in \mathcal{F}$ an *event*. We then say $\mathcal{P}(A)$ the *probability* of A. Then, from this

Definition 3.4: Random Variables

A *random variable* is a function $X : \Omega \to \mathcal{X}$. Then, we say that X has a *distribution* Q, denoted $X \sim Q$, if

$$\mathbb{P}(X \in B) = Q(B)$$

where

$$\mathbb{P}(X \in B) := \mathbb{P}(\{\omega : X(\omega) \in B\})$$

Thus, notice that $Q(B) = \mathbb{P}(X^{-1}(B))$, so Q is the push-forward measure of $\mathbb{P}$ through X.

Thus, we can then say that

$$\mathbb{P}(XY \leq Z^2) = \mathbb{P}(\{\omega : X(\omega)Y(\omega) \leq Z(\omega)^2\})$$

Here's an example to make this more understandable. Let $\Omega = \{1, 2, 3, 4, 5, 6\}^2$, $\mathcal{F}$ be the discrete σ-field, and $\mathcal{P}$ be the counting measure (divided by 36 to normalize). Then, if $X : \Omega \to \{1, 2, 3, 4, 5, 6\}$ is the function that takes the first entry of a pair $(i, j) \in \{1, 2, 3, 4, 5, 6\}^2$, then notice that X induces the uniform distribution on $\{1, 2, 3, 4, 5, 6\}$! If U is the induced measure on $\{1, 2, 3, 4, 5, 6\}$, we see that

$$\mathbb{P}(U = 1) = \mathbb{P}\{(i, j) : i = 1\} = 6/36 = 1/6$$

which is as we expect! Notice that X in this example is the "result of the first die" function on the set of "all possible rolls with two die".

Statistical Models

Definition 3.5: Statistical Models

A *statistical model* $\mathcal{P}$ is a family of candidate probability distributions for a random variable X. That is, we assume $X \sim P$ for some $P \in \mathcal{P}$.

Recall the coin flip example. We will restate it here with this new definition.

Model 1: Assume $X \sim \text{Binom}(n, \theta)$. Then, this means our statistical model is $\mathcal{P} = \{\text{Binom}(n, \theta) : \theta \in [0, 1]\}$. Note that $\text{Binom}(n, \theta)$ has pmf $p_\theta(x) = \binom{n}{x}\theta^x(1-\theta)^{n-x}$, which is a density with respect to the counting measure.

Model 2: Assume $X_i \overset{\text{ind}}{\sim} \text{Binom}(n, \theta_i)$ for $i = 1, \ldots, 48$ with $\theta_i \in [0, 1]$. Then, the parameter space is $[0, 1]^{48}$, instead of just $[0, 1]$ as before.

Model 3: Assume $X_{i,t} \overset{ttind}{\sim} \text{Bern}(\theta_{i,t})$ with $i = 1, \ldots, 48$ and $t = 1, \ldots, n_i$. If we also require that $\theta_{i,1} \geq \theta_{i,2} \geq \cdots \geq \theta_{i,n_i}$, this makes the parameter space smaller, but it is still massive. Because of this, it becomes effectively non-parametric, which we will define now.

Definition 3.6: Parametric Models

A *parametric model* $\mathcal{P}$ is a collection of distributions that are indexed by a parameter $\theta \in \Theta$ (Θ is the parameter space). Typically, $\Theta \subseteq \mathbb{R}^d$, in which case we call d the *model dimension*. We then will use $\mathbb{P}_\theta(\cdot) \in \mathcal{P}$ to be a probability distribution, and $\mathbb{E}_\theta[\cdot]$ for expectation.

Definition 3.7: Non-parametric Models

A *nonparametric model* $\mathcal{P}$ is a collection of distributions that are not naturally indexed by some parameter space. For instance, we can say $\mathcal{P} = \{P : P\text{is a distribution on } \mathbb{R}\}$.

Estimation

Ne question, given $350,000$ flips and $178,000$ heads, what is a good way to estimate θ, the chance that a given flip lands heads.

Bayesian Answer: Assume $\theta \sim U(0,1)$, and use conditionals to find θ that is most likely.

Frequentist Answer: Find a map $E : X \to \Theta$ that maps data to a value for θ, and analyze how good E, called the *estimator* is.

Let's start by analyzing the frequentist methodology. Start with a model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ that we assume X to follow. Suppose we want to guess a value of interest $g(\theta)$ called the *estimand*. The guess we make is $\delta(X)$, called the *estimate*, and the function that generates this estimate, $\delta(\cdot)$ is called the *estimator*.

Now, how do we tell if an estimator is good? We define what is called a *loss function* $L(\theta, d)$, which measures the *disutility* of guessing $g(\theta) = d$. This function is typically nonnegative with the property that $L(\theta, d) = 0 \iff d = g(\theta)$. Typically, we will use the *squared error loss*, which is defined to be $L(\theta, d) = (g(\theta), d)^2$.

Now, this is just a function on a given guess, but in reality we want to measure how well δ does as X varies over it's distribution. This, we introduce the *risk function*, which is defined as

$$R(\theta; \delta(\cdot)) = \mathbb{E}_\theta[L(\theta, \delta(X))]$$

Note that this notation implies we know θ, and we are calculating expectation with respect to the random variable X.

9/8/26 Lecture 4: Estimation and Sufficiency

Recall the setting. We observe data $X \sim P$ for some $P \in \mathcal{P} = \{P_\theta : \theta \in \Theta\}$. We want to guess estimand $g(\theta)$ (which is typically the median, mean, or some coefficient) by applying an estimator $\delta(\cdot)$ to X.

A good guess is something that minimizes a function $L(\theta, d)$, which is called the loss. This measures the disutility of a guess. Typically, we take $L(\theta, d) = (g(\theta) - d)^2$. Finally, to tell how good a estimator is given any sample, we use the risk function

$$\begin{aligned} R(\theta; \delta) &= \mathbb{E}_\theta[L(\theta, \delta(X))] \\ &= \int_X L(\theta, \delta(X)) dP_\theta(X) \end{aligned}$$

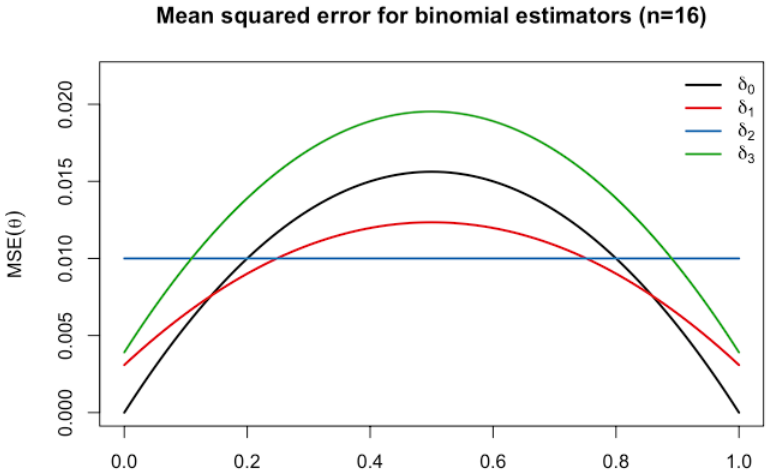
for example, an example of the risk function is

$$\text{MSE}(\delta) = \mathbb{E}_\theta[(g(\theta) - \delta(X))^2]$$

For instance, consider $X \sim \text{Binom}(n, \theta)$, which has probability mass function $p_\theta(x) = \binom{n}{x}\theta^x(1-\theta)^{n-x}$. We want to estimate $g(\theta) = \theta$. Consider the following estimators

$$\delta_0(X) = \frac{X}{n}, \quad \delta_1(X) = \frac{X+1}{n+2}, \quad \delta_2(X) = \frac{X+2}{n+4}, \quad \text{and } \delta_3(X) = \frac{X+1}{n}$$

Notice that although the first one is unbiased, the others have smaller variance, so their MSEs are comparable. Here is a graph of their MSEs for a given θ.



We see that between $\theta_0, \theta_1, \theta_2$, there is always one θ that makes one estimator better than another. However, we see that θ_3 is strictly worse than θ_0, which means there is no reason to use it over θ_0. We will soon formalize what this means, but first: Why don't we just consider an estimator that is the best among all estimators?

Well, the reason is that it doesn't exist. If there exists an estimator that is always better than every estimator everywhere, then for a given θ, we can take an estimator that simply gives the constant estimate θ. Then, it has no MSE, which implies that our "best everywhere" estimator must have no MSE everywhere, which simply cannot happen.

Definition 4.1: Inadmissable Estimators

We say an estimator δ is *inadmissable* if there exists an estimator δ^* such that

1. $R(\theta;\delta^*) \leq R(\theta;\delta)$ for all $\theta \in \Theta$.
2. $R(\theta;\delta^*) < R(\theta;\delta)$ for some $\theta \in \Theta$.

We then say that δ^* strictly dominates δ.

Thus, in our case above, δ_3 is inadmissable.

Now, given multiple estimators that are inadmissable, how do we remove the ambiguity. The way to do this is to reduce the risk function $R(\theta;\delta)$ to a scalar (not dependent on θ). There are multiple ways to do this.

Method 1: Average Case Risk: This involes measuring the average case risk, which means we are going to average over θ. Thus, we need to assume that $\theta \sim \pi$ and minimize $\int_\theta R(\theta;\delta)d\pi(\theta)$. If π is a probability measure, then we are trying to find $\mathbb{E}_\pi[R(\theta;\delta)]$. For instance, if π is uniform on $[0,1]$ in the case above, then we summarize the risk functions by integrating over the MSE over $[0,1]$.

Notice that this isn't inherently Bayes. We aren't assuming that θ takes a distribution, but we are using this distribution as a way to average θ. However, if we are doing Bayesian statistics, then this would be the way to pick the best estimator.

Method 2: Worse Case Risk: With this method, we are trying to minimize, over θ, the $\sup_{\theta\in\Theta} R(\theta;\delta)$. From out example above δ_2 minimizes worse case risk among the other estimators.

Method 3: Restrict Estimators: One thing we can do is to just rule out all unbiased estimators and try Method 1 or Method 2 on the resulting list. This has merit in some cases, but we will see sometimes it's useful to use biased estimators.

Sufficiency

Recall the three models of coin flips:

Model 3: $X_{i,t} \overset{\text{ind}}{\sim} \text{Bern}(\theta_{i,t})$, for $i \leq 48$ and $t \leq n_i$

Model 2: $X_i \overset{\text{ind}}{\sim} \text{Binom}(\theta_i)$ for $i \leq 48$. Notice that $X_i \sum_{t=1}^{n_i} X_{i,t}$ where $X_{i,t} \overset{\text{ind}}{\sim} \text{Bern}(\theta_i)$.

Model 1: $X \overset{\text{ind}}{\sim} \text{Binom}(n^+,\theta)$. Notice that $X = \sum_i X_i = \sum_{i,t} X_{i,t}$.
Thus, in some sense

$$\text{Model 1} \subseteq \text{Model 2} \subseteq \text{Model 3}$$

as given that we know $X_{i,t}$ but Model 2 is our assumption, we can construct X_i, and given that we know X_i but Model 1 is our assumption, we can construct X.

Definition 4.2: Sufficient Statistics

A statistics $T(X)$ is *sufficient* for a model $\mathcal{P}$ if the conditional distribution of $X|T(X)$ is the same for all $P \in \mathcal{P}$. In other words, no matter what $\mathcal{P}$ is, knowing that $T(X) = c$ can tell you what the distribution of X is.

For example, if $X_1,\ldots,X_n$ are i.i.d. $\text{Bern}(\theta)$, then $T(X) = \sum_i X_i$ is sufficient, because knowing that $T(X) = c$ can give you the marginal distribution of $X = (X_1,\ldots,X_n)$.

Now, notice that X_i is sufficient for $\text{Binom}(n_i, \theta_i)$, it is not sufficient for $\text{Bern}(\theta_{i,t})$, and is not minimal sufficient for $\text{Binom}(n, \theta)$ (as it gives too much information).

Factorization Theorem

Theorem 4.3: Factorization Theorem

Let $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ be a model with densities $p_\theta(X)$ with respect to a common dominating measure μ on X. Then, $T(X)$ is sufficient if, and only if, there exists a $g_\theta(t)$ and $h(x) \geq 0$ such that

$$p_\theta(x) \overset{\mu \text{ a.e.}}{=} g_\theta(T(x)) \cdot h(x)$$

Proof: We will prove it for discrete X, as the proof for the continuous case is similar but just requires measure theoretic notions. First, assume wlog that $\mu = \#$ (as every measure is absolutely continuous with respect to $\#$).

($\Leftarrow$) We see that

$$\begin{aligned}
\mathbb{P}_\theta(X = x \mid T = t) &= \frac{\mathbb{P}_\theta(X = x, T(x) = t)}{\mathbb{P}_\theta(T(x) = t)} \\
&= \frac{\mathbb{P}_\theta(X = x, T(x) = t)}{\sum_{z\,:\,Tz = t} \mathbb{P}_\theta(X = z)} \\
&= \frac{g_\theta(t)h(x) \cdot 1\{T(x) = t\}}{\sum_{T(z)=t} g_\theta(t)h(z)} \\
&= \frac{h(z)1\{T(x) = t\}}{\sum_{T(z)=t} h(z)}
\end{aligned}$$

Thus, we see that this doesn't depend on θ, so is the same for all $P \in \mathcal{P}$.

($\Rightarrow$) Assume that T is sufficient. Then, let $g_\theta(t) = \mathbb{P}_\theta(T(X) = t)$ and $h(x) = \mathbb{P}_\theta(X = x \mid T(x) = t)$. Then, as

$$\begin{aligned}
g_\theta(T(x))h(x) &= \mathbb{P}_\theta(T(X) = T(x) \text{ and } X = x) \\
&= \mathbb{P}_\theta(X = x) \\
&= p_\theta(x)
\end{aligned}$$

which was to be shown.

■

Note: We can absorb $h(x)$ into μ by letting ν such that $\frac{d\nu}{d\mu} = h$. Thus, we would then have $p_\theta(X) = g_\theta(T(X))$, which is what we kind of mean by "T gives the distribution of X".

For example, if $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\theta, 1)$, then as the density function is $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-\theta)^2}$, we see that

$$\begin{aligned}
p_\theta(x) &= (2\pi)^{-n/2} \prod_{i=1}^{n} e^{-1/2(x_i - \theta)^2} \\
&= e^{\theta \sum_{i=1}^n x_i - n\theta^2/2} \cdot \frac{e^{-\sum x_i^2/2}}{(2\pi)^{n/2}}
\end{aligned}$$

which is the desired factorization showing that $\sum_{i=1}^n x_i$ is sufficient.

Another example is that if $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Pois}(\theta)$, then as $\phi(x) = \frac{\theta^x e^{-\theta}}{x!}$ for $x = 0, 1, \ldots$, then as

$$p_\theta(x) = \prod_{i=1}^n \frac{\theta^{x_i} e^{-\theta}}{x_i!} = \theta^{\sum x_i} e^{-n\theta} \cdot \frac{1}{\prod_{i=1}^n x_i!}$$

we see that $\sum_{i=1}^x x_i$ is sufficient.

One last interesting example is that, if $X_i \overset{\text{iid}}{\sim} U[\theta, \theta + 1]$, then as $\theta(x) = 1\{\theta \le x \le \theta + 1\}$, we see that

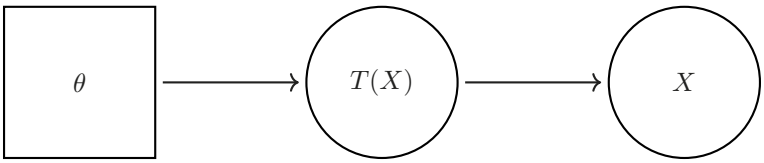
$$p_\theta(x) = \prod_{i=1}^n 1\{\theta \le x_i \le \theta + 1\} = 1\{\theta \le \min_i X_i \le \max_i X_i \le \theta + 1\}$$

we can conclude that $(\min_i X_i, \max_i X_i)$ is sufficient.

9/10/26 Lecture 5: More on Sufficiency

Recall that if $X_1, X_2 \overset{\text{iid}}{\sim} \text{Binom}(n, \theta)$, then $X_1 + X_2$ is a sufficient statistics. However, if $X_1, X_2 \overset{\text{iid}}{\sim} e^{-|x-n\theta|}$, then it turns out $X_1 + X_2$ aren't sufficient statistics.

However, in the above two cases, both $(X_{(1)}, X_{(2)})$ are sufficient. Notice that this is due to the fact they are independent and identically distributed, so that are exchangeable (just rename one to the other). Thus, we can conclude that if X_1, X_2 are $\overset{\text{iid}}{\sim}$, then $(X_{(1)}, X_{(2)})$ is sufficient! Consider the following diagram for a sufficient statistic T.



That is, knowing θ tells you the distribution of $T(X)$, and knowing $T(X)$ tells you the distribution of X. Thus, in a sense, knowing all of X is pointless, we just need to know $T(X)$.

A funny but effective way of remembering this is that, if a group of scientists give you the data X, and you compute $T(X)$ and accidentally throw out X, then if the scientists ask you for the data you can generate your own sample $\tilde{X}$ with $T(X)$ and no one would ever be able to tell you lost your data.

What this also means is that if we have a sufficient statistics T, then we should have our estimators be functions of T as using X just introduces unnecessary randomness from the conditional distribution of X given T (which we would know). This is what is called a *sufficiency reduction*: just using $T(X)$ instead of X.

Order Statistics

Definition 5.1: Order Statistics

For $X = (X_1, X_2, \ldots, X_n) \in \mathbb{R}^n$, define the *order statistics* of X to be

$$\min_i X_i = X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)} = \max_i X_i$$

where $\{X_{(1)}, \ldots, X_{(n)}\} = \{x_1, \ldots, x_n\}$.

Now, if $X_1, \ldots, X_n \overset{\text{iid}}{\sim} P_\theta$, and $\mathcal{P} = \{P_\theta^n : \theta \in \Theta\}$ is any model, then due to the invariance of $X_1, \ldots, X_n$ under permutations, this immediately implies the statistics $(X_{(i)})_{i=1}^n = (X_{(1)}, \ldots, X_{(n)})$ is sufficient. Thus, there is a correspondence

$$(X_{(i)}) \text{ is sufficient} \iff \text{model is symmetric}$$

Empirical Distribution

Definition 5.2: The Dirac Measure

Define the *Dirac measure* to be $\delta_x(A) = 1\{x \in A\}$ for some $x \in A$. The *empirical distribution* is, given random variables, $X_1, \dots, X_n$

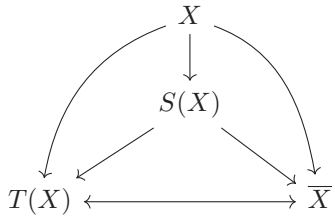
$$\hat{P}_n(\cdot) = \frac{1}{n}\sum_{i=1}^{n} \delta_{X_i}(\cdot)$$

so $\hat{P}_n(A) = \#\{i : X_i \in A\}/n$. Notice that $\hat{P}_n(\cdot)$ is a random measure, depending on the $X_1, \dots, X_n$.

Given iid $X_1, \dots, X_n$, since the multiset $\{X_1, \dots, X_n\}$ is sufficient, we see that $\hat{P}_n$, which tells you the multiset, is also sufficient.

Now, lets move on to minimal sufficiency. Notice that if $X_1, X_2 \overset{\text{iid}}{\sim} \text{Binom}(n, \theta)$, then $X = (X_1, X_2)$ is sufficient, $(X_{(1)}, X_{(2)})$ is sufficient, and so is $X_1 + X_2$.

However, notice that given (X_1, X_2), we can generate $S(X) = (X_{(1)}, X_{(2)})$, and given $(X_{(1)}, X_{(2)})$, we can generate $T(X) = X_1 + X_2$. Thus, this gives us a diagram



where $\overline{X} = \frac{X_1+X_2}{2}$. The arrows that connect two sufficient statistics tells us that we can compute one statistic from the other. That is, $A \to B$ implies $B = f(A)$. Thus, this allows us to define what a minimally sufficient statistic is. First, a quick proposition:

Proposition 5.3

If $T(X)$ is sufficient and $T(X) = f(S(X))$, then $S(X)$ is sufficient.

Proof: We use the factorization theorem, noting

$$p_\theta(x) = g_\theta(T(x))h(x) = (g_\theta \circ f)(S(x))h(x)$$

which tells us that $S(X)$ is sufficient. ■

Definition 5.4: Minimal Sufficient Statistic

A sufficient statistic $T(X)$ is *minimal sufficient* if

1. $T(X)$ is sufficient.
2. If $S(X)$ is sufficient, then $T(X) = f(S(X))$ for some f.

Bayesian Interpretation of Sufficiency

Consider a model $\mathcal{P} = \{p_\theta : \theta \in \Theta\}$, a prior $\theta \sim \pi$, and data $x \mid \theta \sim p_\theta$. Then, we see that

$$\pi(\theta \mid x) = \frac{\pi(\theta)p_\theta(x)}{\int \pi(\gamma)p_\gamma(x)d\gamma}$$

then, if T is sufficient, we see that

$$\begin{aligned}\frac{\pi(\theta)p_\theta(x)}{\int \pi(\gamma)p_\gamma(x)d\gamma} &= \frac{\pi(\theta)g_\theta(T(x))h(x)}{\int_\gamma \pi(\gamma)g_\gamma(T(x))h(x)d\gamma} \\ &= \frac{\pi(\theta)g_\theta(T(x))}{\int \pi(\gamma)g_\gamma(T(x))d\gamma}\end{aligned}$$

Thus, we see that the value of $T(x)$ determines our posteriors $\pi(\theta \mid x)$. Thus, what sufficiency does in this case is collapsing what are "equivalent" (give the same posterior) data sets.

Now, we know that $\pi(\theta \mid x) \propto_\theta \pi(\theta)p_\theta(x)$, which implies that $\pi(\theta \mid x) = \pi(\theta \mid y)$ if, and only if, $\frac{p_\theta(x)}{p_\theta(y)}$ is constant in θ. Thus, changing θ does not change this likelihood ratio, which means that if x and y have the same sufficient statistic, this ratio is constant over θ.

Now, heuristically, we see that a typical sufficient statistic collapses data, but it doesn't maximally collapse data, which is what a maximal sufficient statistic would do. This, along with the above insight can tell us a necessary and sufficient condition for minimally sufficient statistics.

First, assuming that $\mathcal{P}$ has densities p_θ on a sample space X, we can define an equivalence relation

$$x \equiv_p y \text{ iff } \frac{p_\theta(x)}{p_\theta(y)} \text{ does not depend on } \theta$$

First, notice that if $T(x) = T(y)$, then $x \equiv_p y$ from above. Now, consider the map C that sends x to it's equivalence class under $\equiv_p$. Under some mild conditions, we can show that this is a sufficient statistics. Also, as $T(x) = T(y) \implies x \sim_p y \implies C(x) = C(y)$, we see that we can write $C(x) = f(T(x))$. As this holds for all sufficient statistics, this implies that C is minimal!

Theorem 5.5 (Bahader)

$T(X)$ is minimal sufficient if $x \equiv_p y \iff T(x) = T(y)$. That is, the equivalence relation induced by T is $\equiv_p$.

Now, lets see how we can find these minimal sufficient statistics.

Definition 5.6: Likelihood Functions

Assume $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ has densities p_θ. The *likelihood function* is $\text{Lik}(\theta; X) = p_\theta(X)$, which is a random function of θ. The *log-likelihood function* is $\ell(\theta; X) = \log p_\theta(X)$.

Now, notice that as

$$x \equiv_\theta y \iff \ell(\theta; x) - \ell(\theta; y) \text{ is constant in } \theta$$

Thus, we see that the equivalence classes of $\ell(\theta; X)$ up to scaling are minimal sufficient by Theorem 5.5! What this also means is that $T(X)$ is minimal sufficient if $\ell(\theta; x) - \ell(\theta; y)$ is constant in θ if, and only if, $T(x) = T(y)$.

9/15/26 Lecture 6: Exponential Families

Exponential Families

Definition 6.1: Exponential Families

An *s-parameter exponential family* is a family $\mathcal{P} = \{P_\eta : \eta \in \Xi\}$ with densities of the form

$$p_\eta(x) = e^{\eta' T(x) - A(\eta)} h(x)$$

with respect to a common dominating measure μ on X. Recall that $\eta' := \eta^\top$.

Each of the components in the density has a name:

- $\eta \in \Xi \subseteq \mathbb{R}^s$ is called the *natural parameter*
- $T : \chi \to \mathbb{R}^s$ is the s-dimensional *sufficient statistic*
- $h : \chi \to [0, \infty)$ is called the *base-density* or *carrier-density*
- $A : \Xi \to \mathbb{R}$ is called the *log-partition function* or *normalizing constant*

Now that, as p_η must be a probability density, we can solve for $A(\eta)$ to get

$$A(\eta) = \log \int_\chi e^{\eta' T(x)} h(x) d\mu \leq \infty$$

Notice that sometimes this can be infinity, which is not good, so we give the following definition.

Definition 6.2: Natural Parameter Space

The *natural parameter space* is $\Xi_1 = \{\eta : A(\eta) < \infty\}$. That is, the set of η for which $A(\eta)$ is finite.

It can be shown that (using Hölder's inequality!) as the function $A(\eta)$ is convex, we also know Ξ_1 is a convex set, which may come in handy.

Heuristically: The reason we care about these families is that they are in some sense the closest we can get to a "hyperplane" of densities. The reason is that, absorbing $h(x)$ into μ, we get $p_\eta(x) = e^{\eta' T(x) - A(\eta)}$, so

$$\log p_\eta(x) = \eta' T(x) - A(\eta)$$

removing the normalizing factor, we get a hyperplane! Now, there is still the log in front, but this turns out to be more of a blessing in the curse, because logs turn multiplication into addition, so we are, in a sense, linearizing multplication which is good since we do a lot more multiplication than addition in statistics.

Exapmle 1: Poisson

Let $X \sim \text{Pois}(\lambda)$ which has pmf $\frac{\lambda^x e^{-\lambda}}{x!}$. Then, we can write

$$p_\lambda(x) = e^{x \log \lambda - \lambda} \cdot \frac{1}{x!}$$

Now, to put it in the form of the exponential family above, we must then have that $\eta(\lambda) = \log(\lambda)$, $T(x) = x$, $A(\eta) = \lambda = e^\eta$, and $h(x) = \frac{1}{x!}$, so that

$$p_\eta(x) = e^{\eta x - e^\eta} \frac{1}{x!}$$

A question one may have is that: Why didn't we just leave it in terms of λ? Well, having it explicitly in this form will prove itself useful soon.

Example 2: Poisson, n Variables

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Pois}(\lambda)$. We see that

$$\begin{aligned} p_\lambda(X) &= \prod_{i=1}^n e^{\log \lambda x_i - \lambda} \frac{1}{x_i!} \\ &= e^{\log \lambda \sum_{i=1}^n x_i - n\lambda} \cdot \frac{1}{\prod_{i=1}^n x_i!} \end{aligned}$$

Thus, in this case $\eta = \log \lambda$, $T(x) = \sum_{i=1}^n x_i$, $A(\eta) = ne^\eta$, and $h(x) = \frac{1}{\prod_{i=1}^n x_i!}$

Example 3: Generic, n Variables

Suppose that

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} p_{eta}^{(1)} = e^{\eta' T^{(1)}(x) - A^{(1)}(\eta)} \cdot h^{(1)}(x)$$

where the (1) means each density is for a single variable. Then,

$$\begin{aligned} p_\eta(x) &= \prod_{i=1}^n e^{\eta' T^{(1)}(x_i) - A^{(1)}(\eta)} h^{(1)}(x) \\ &= e^{\eta' \sum_{i=1}^n T^{(1)}(x_i) - nA^{(1)}(\eta)} \prod_{i=1}^n h^{(1)}(x_i) \end{aligned}$$

Thus, we see that $\eta = \eta$, $T(x) = \sum_{i=1}^n T^{(1)}(x_i)$, $A(\eta) = nA^{(1)}(\eta)$, and $h(x) = \prod_{i=1}^n h^{(1)}(x_i)$.

Differential Identites

Notice that on Ξ_1

$$e^{A(\eta)} = \int e^{\eta' T(x)} h(x) d\mu$$

Let $g(\eta) = \int f(x) e^{\eta' T(x)} h(x) d\mu$. On the interior of where $\int \left| f(x) e^{\eta' T(x)} h(x) \right| d\mu$ is finite, $g(\eta)$ is infinitely differentiable (this is via a theorem from real analysis), so on the interior of Ξ_1, we see that $e^{A(\eta)}$ is infinitely differentiable. Thus, taking the derivative with respect to a component η_j, we see that, assuming we can differentiate under the integral sign.

$$\begin{aligned} \frac{\partial}{\partial \eta_j} e^{A(\eta)} &= \int \frac{\partial}{\partial \eta_j} e^{\eta' T(x)} h(x) d\mu \\ \frac{\partial A}{\partial \eta_j}(\eta) e^{A(\eta)} &= \int T_j(x) e^{\eta' T(x)} h(x) d\mu \\ \frac{\partial A}{\partial \eta_j}(\eta) &= \int T_j(x) e^{\eta' T(x) - A(\eta)} h(x) d\mu \\ &= \mathbb{E}_\eta[T_j(X)] \end{aligned}$$

Thus, as this holds for all partial derivatives, we see that $\nabla_\eta A(\eta) = \mathbb{E}_\eta [T(X)]$. That's a nice way to compute the expectation, but can we get more?

Taking the second derivative with respect to another component η_k (which may be the same as before), we see

$$\begin{aligned}
\frac{\partial^2}{\partial\eta_j\partial\eta_k} e^{A(\eta)} &= \int \frac{\partial^2}{\partial\eta_j\partial\eta_k} e^{\eta' T(x)} h(x) d\mu \\
e^{A(\mu)} \left(\frac{\partial^2 A}{\partial\eta_j\partial\eta_k}(\eta) + \frac{\partial A}{\partial\eta_j}(\eta)\frac{\partial A}{\partial\eta_j}(\eta) \right) &= \int T_j(x) T_k(x) e^{\eta' T(x)} h(x) d\mu \\
\left(\frac{\partial^2 A}{\partial\eta_j\partial\eta_k}(\eta) + \frac{\partial A}{\partial\eta_j}(\eta)\frac{\partial A}{\partial\eta_j}(\eta) \right) &= \int T_j(x) t_k(x) e^{\eta' T(x) - A(\eta)} h(x) d\mu \\
\left(\frac{\partial^2 A}{\partial\eta_j\partial\eta_k}(\eta) + \mathbb{E}_\eta[T_j(X)]\mathbb{E}_\eta[T_k(X)] \right) &= \mathbb{E}_\eta[T_j(X)T_k(X)] \\
\frac{\partial^2 A}{\partial\eta_j\partial\eta_k}(\eta) &= \mathrm{Cov}_\eta(T_j(X), T_k(X))
\end{aligned}$$

Thus, we see that $\nabla^2_\eta A(\eta) = \mathrm{Var}_\eta(T(X)) \in \mathbb{R}^{s\times s}$, where ∇^2_η is the Hessian, and $\mathrm{Var}_\eta(T(X))$ is the variance-covariance matrix.

Example 4: Poisson

Recall from above we have that $T(x) = x$ and $A(\eta) = e^\eta = \lambda$. Then, we see that

$$\mathbb{E}_\eta[X] = \frac{dA}{d\eta} = e^\eta = \lambda$$

and

$$\mathrm{Var}_\eta[X] = \frac{d^2 A}{d\eta^2} = e^\eta = \lambda$$

which is expected! Notice that if we were to not replace η with λ as we did above, and had $A(\lambda) = \lambda$, then we would see that

$$\mathbb{E}_\eta[X] = 1 \quad \text{and} \quad \mathrm{Var}_\eta[X] = 0$$

which is wrong.

Moment Generating Functions

Now, because taking derivatives gives moments, there must be some connection to moments right? Here it is:

$$\begin{aligned} M_\eta^{T(X)}(u) &= \mathbb{E}_\eta[e^{u'T(x)}] \\ &= \int e^{u'T(x)} e^{\eta' T(x) - A(\eta)} h(x) d\mu \\ &= \frac{e^{A(\eta+u)}}{e^{A(\eta)}} \int e^{(u+\eta)'T(x) - A(u+\eta)} h(x) d\mu(x) \\ &= e^{A(\eta+u)-A(\eta)} \end{aligned}$$

where the third step follows from the fact we are integrating a probability density over the space, which requires that $u + \eta \in \xi_1$, but forget about it.

Anyways, taking derivatives with respect to u gives

$$\nabla_u e^{A(\eta+u)-A(\eta)} = \nabla(A(\eta+u)) e^{A(\eta+u)-A(\eta)}$$

Then, as $u = 0$, we get

$$\nabla_u M_\eta^{T(X)}(u) = \nabla(A(\eta)) = \mathbb{E}_\eta[T(X)]$$

which is as expected!

There is also another related function called the cumulant-generating function, which looks like

$$K_\eta^{T(X)}(u) = \log M_\eta^{T(X)}(u) = A(\eta+u) - A(\eta)$$

but we will talk about these later.

9/17/26 Lecture 7: Exponential Families

Recall, an exponential family looks like $\mathcal{P} = \{p_\eta : \eta \in \Xi\}$ with

$$p_\eta(x) = e^{\eta' T(x) - A(\eta)} h(x)$$

These give nice forms for moments, which are the differential identities

$$\nabla A(\eta) = \mathbb{E}_\eta[T(X)], \quad \nabla^2 A(\eta) = \mathrm{Var}_\eta(T(X)), M_\eta^{T(X)}(u) = e^{A(\eta+u)-A(\eta)}$$

The reason these are nice is that a product of exponential family is an exponential family, so these moments are easy to compute.

Also, one last note, we may also find it convenient to parameterize our exponential families in a different form, like

$$p_\eta(x) = e^{\eta(\theta)' T(x) - B(\theta)} h(x)$$

where $B(\theta) = A(\eta(\theta))$, so we may have to "massage" these forms into the canonical form. Here are some examples.

Example

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ unknown. Then, we see that when $\theta = (\mu, \sigma)$, we have

$$p_\theta(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

but by expanding, we can write this like

$$p_\theta(x) = e^{\frac{\mu}{\sigma^2}x - \frac{x^2}{2\sigma^2} - \frac{\mu^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)}$$

In this case, we see that

$$\eta(\theta) = \begin{bmatrix} \frac{\mu}{\sigma^2} \\ -\frac{1}{2\sigma^2} \end{bmatrix}, \quad T(x) = \begin{bmatrix} x \\ x^2 \end{bmatrix}, \quad B(\theta) = \frac{\mu^2}{2\sigma^2} + \frac{1}{2}\log(2\pi\sigma^2), \quad h(x) = 1$$

where we can put it back in canonical form writing

$$A(\eta(x)) = \frac{-\eta_1^2}{4\eta_2} + \frac{1}{2}\log(-\frac{\pi}{\eta_2})$$

so

$$p_\eta(x) = e^{\eta' \begin{pmatrix} x \\ x^2 \end{pmatrix} - A(\eta)} h(x)$$

Notice something interesting. $T(X)$, the sufficient statistic, was supposed to be a data reduction, but now we turned it into a vector that has arguably (not really) more data! Where this helps however is when you have multiple variables; then, the sufficient statistic is still just a vector whereas the data is a n-tuple.

Example

Let $X \sim \mathrm{Binom}(n, \theta)$ with θ unknown, then, we can write

$$p_\theta(x) = \theta^x (1-\theta)^{n-x} \binom{n}{x}$$

Now, it seems like we will need a sufficient statistic consisting of x and $n-x$, but notice that these have a very intimate relation; their sum must be n! Thus, we can reduce the dimension by writing

$$\begin{aligned} p_\theta(x) &= \left(\frac{\theta}{1-\theta}\right)^x (1-\theta)^n \binom{n}{x} \\ &= e^{x \log \frac{\theta}{1-\theta} - n \log \frac{1}{1-\theta}} \binom{n}{x} \end{aligned}$$

so we have

$$\eta = \log \frac{\theta}{1-\theta}, \quad x = T(x), \quad A(\eta) = n \log \frac{1}{1-\theta}, \quad h(x) = \binom{n}{x}$$

One way we can think of exponential families is as "exponential tilting" of the carrier $h(x)$. That is, we start with our carrier distribution $h(x)$, then we tilt by $e^{\eta' T(x)}$, and then we renormalize using $e^{-A(\eta)}$. What this does is that it tilts the density towards points for which $\eta' T(x)$ is large, and away from points where $\eta' T(x)$ is small.

For example, consider the exponential family

$$p_\eta(x) \propto e^{\eta_1 x + \eta_2 x^2 + \eta_3 \cos(100x)}(1 - |x|)$$

Increasing η_1 pushes the distribution to the right, increasing η_2 pushing the distribution to the edges, and increasing η_3 makes p_η oscillate "like cosine".

Also, note that decomposition of an exponential family into η, T, A, h is very not unique. Firstly, it only matters what the span of $T(x)$ is, as choosing η allows us to map any element of the span of T to another. Additionally, we can always add constants to the terms in $T(x)$, because that can be taken out and absorbed into the $A(\eta)$ term.

Now, we will prove a quick result which will reinforce the notion that the sufficient statistic is what we should care about.

Proposition 7.1

Suppose $X \sim p_\eta(x) = e^{\eta' T(x) - A(\eta)}$ with respect to μ. Then,

$$T(x) \sim q_\eta(t) = e^{\eta' t - A(\eta)}$$

with respect to $\nu = \mu \circ T^{-1}$.

Proof: We see that

$$\begin{aligned} \mathbb{P}_\eta(T(X) \in B) &= \int_\chi 1_B(T(x)) e^{\eta' T(x) - A(\eta)} d\mu(x) \\ &= \int_t 1_B(t) e^{\mu' t - A(\eta)} d\nu(t) \end{aligned}$$

where we are basically doing a change of coordinates, $x \to T(x)$. This can be easier seen with the counting measure

$$\sum_{x\,:\,T(x)\in B} e^{\eta' T(x) - A(\eta)} \mu(\{x\}) = \sum_{t \in B} e^{\eta' t - A(\eta)} \mu(T^{-1}(t))$$

In words, $p_\eta(x)$ is a density for each x, $q_\eta(t)$ is a density for each slice $\{x : T(x) = t\}$.

Through these lens, exponential tilt is taking a carrier density and tilting it on the slices $\{x : T(x) = t\}$. This makes sense to do because tilting by $e^{\eta' T(x)}$ does not depend on x, but on where T sends x. Also, this is nice because if X is an exponential family, $T(X)$ is also an exponential family, but it actually is of a special kind.

Definition 7.2: Canonical Form

If the densities of our exponential family are of the form

$$p_\eta(x) = e^{\eta' x - A(\eta)}$$

we say this exponential family is in *canonical form.*

From the remarks above, notice we can always put an exponential family in canonical form by absorbing h into our measure, and performing a sufficiency reduction $x \to T(x)$.

Minimal Form

Definition 7.3: Minimal Form

We say an exponential family

$$p_\eta(x) = e^{\eta' T(x) - A(\eta)} h(x)$$

is in *minimal form* if $\eta \in \Xi$ and $T(x)$ satisfy no linear constraints. That is, there are no $a \neq 0$ and $b \in \mathbb{R}$ such that $\eta' a = b$ or $T(x)' a = b$ $\mathcal{P}$ almost surely. (where $\mathcal{P}$ almost surely means this holds on a set of full measure for some η).

The reason we care about minimal form is because, if an exponential family is not in minimal form, we can use the linear constrains to reduce the dimension from a s-dimensional exponential form to an r-dimensional exponential form with $r < s$.

Lemma 7.4

If p_η is minimal (in minimal form), then $T(x)$ is minimal sufficient.

Proof: We want to show that if $\ell(\cdot, x) = \ell(\cdot, y)$ up to scalar, then $T(x) = T(y)$. Notice that since T is sufficient, the converse of this statement holds.

First, we see that

$$\ell(\eta, x) - \ell(\eta, y) = \eta'(T(x) - T(y)) + \log(h(x)) - \log(h(y))$$

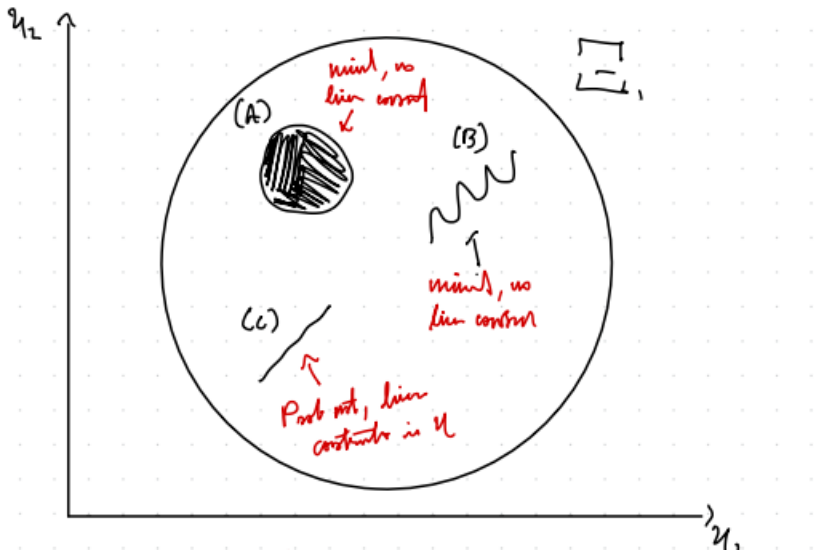
letting $a = T(x) - T(y)$, then if $\ell(\eta, x) - \ell(\eta, y) = C$ for a constant C, then we see that

$$C - \log(h(x) - \log(h(y))) = \eta' a$$

Then, as the LHS is constant in η, then if $a \neq 0$, this contradicts the fact that p_η is minimal. Thus, we must have that $a = 0$, so $T(x) = T(y)$.

■

Consider the following diagram



where $(A), (B)$ and (C) are exponential families living inside Ξ_1. Which ones are minimal? Firstly, we can definitely see that A and B are not minimal, because there is no way to write the η that belong to those families as a solution to a linear equation. However, for (C), we can.

that is, what we can do is pick a base point η_0 on that line, and for any other η on that line, we can write $\eta = \eta_0 + \theta\gamma$ for γ the unit vector in the direction of the line. Then, we see that we have

$$\begin{aligned} e^{\eta' T(x) - A(\eta)} h(x) &= e^{(\eta_0 + \theta\gamma)' T(x) - A(\eta_0 + \theta\gamma)} h(x) \\ &= e^{\theta \cdot \gamma' T(x) - A(\eta_0 + \theta\gamma)} e^{\eta_0' T(x)} h(x) \end{aligned}$$

Thus, our new, lower dimensional natural parameter is θ, our new sufficient statistic is $\gamma' T(x)$, and our new normalizing factor is $A(\eta_0 + \theta\gamma) = B(\theta)$. We also have $h(x) = e^{\eta_0' T(x)} h(x)$.

Also, interestingly, on B which is a 1-dimensional manifold, we can locally linearize so if we can restrict our family in question to a small subset, we see that it can be locally minimal in a sense.

Completeness

Definition 7.5: Completeness of a Sufficient Statistic

We say a sufficient statistic $T(X)$ is *complete* for the family $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ if $\mathbb{E}_\theta[f(T(X))] = 0$ for all θ implies that $f(T(X)) \overset{\text{a.s.}}{=} 0$.

Example

Consider the Laplace location family with minimal sufficient statistic $S(X) = (X_{(1)}, \ldots, X_{(n)})$ with $X_i \overset{\text{iid}}{\sim} \text{Lap}(\theta)$. Is $S(X)$ complete?

The answer is no! Notice that $\mathbb{E}_\theta[X_{(n)} - X_{(1)}] = \mathbb{E}_\theta[Z_{(n)} - Z_{(1)}]$ where $Z_i = X_i - \theta \sim \text{Lap}(0)$. Thus, we see that for

$$f(S(X)) = (X_{(n)} - X_{(1)}) - \mathbb{E}[Z_{(n)} - Z_{(1)}]$$

we have that

$$\mathbb{E}(f(S(X))) = 0$$

but $f(S(X)) \neq 0$ for all θ!

What this means is that $S(X)$, despite being sufficient, still has a lot of "fluff". It gives more information than needed in a sense.

9/22/26 Lecture 8: Complete Statistics

Recall the definition.

Definition 8.1: Completeness

$T(X)$ is *complete* for $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ if

$$\mathbb{E}_\theta[f(T(X))] = 0 \forall \theta \implies f(T(x)) \overset{\text{a.s.}}{=} 0$$

That is, no nontrivial functions of T have mean zero.

Question: What if $\mathbb{E}_\theta[\delta_1(X)] = \mathbb{E}_\theta[\delta_2(X)] = g(\theta)$ for all θ and X is complete?

Answer: This implies that $\delta_1 \overset{\text{a.s.}}{=} \delta_2$, as we see that

$$\mathbb{E}_\theta[(\delta_1 - \delta_2)(X)] = 0$$

Similarly, if we had two functions of the complete statistic whose mean is the same, then they must be the same almost everywhere. We can then reason that if a sufficient statistic is complete, then there is only one unbiased estimator up to almost everywhere equivalence.

Example: German Tank Problem

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} U[0, \theta]$, $\theta \in (0, \infty)$. Then,

$$p_\theta(x) = \prod_{i=1}^n \frac{1}{\theta} 1\{x_i \le \theta\} = \frac{1}{\theta^n} 1\{x_{(n)} \le \theta\}$$

By the factorization theorem, we see that $x_{(n)}$ is a sufficient statistic, and it must be minimal sufficient as clearly $x \equiv_p y \implies x_{(n)} = y_{(n)}$.

Now, we want to find the density of $T(X) = X_{(n)}$. We see

$$\mathbb{P}_\theta(T \le t) = \left(\frac{t}{\theta} \wedge 1\right)^n = \frac{t^n}{\theta^n} \wedge 1$$

Then,

$$p_\theta(t) = \frac{d}{dt}\mathbb{P}_\theta(T \le t) = n\frac{t^{n-1}}{\theta^n} 1\{t \le \theta\}$$

Now, suppose that we have an $f(t)$ for which

$$\mathbb{E}_\theta[f(T(X))] = 0 \quad \forall \theta$$

This implies that

$$\begin{aligned} 0 &= \int_0^\theta f(t) p_\theta(t) dt \\ &= \frac{n}{\theta} \int_0^\theta f(t) t^{n-1} dt \\ \implies &\int_0^\theta f(t) t^{n-1} dt = 0 \quad \forall \theta \end{aligned}$$

Differentiating with respect to θ, we see that

$$0 = f(\theta)\theta^{n-1}$$

which implies that $f(\theta) = 0$. Thus, we see that $T(X) = X_{(n)}$ is complete.

■

Definition 8.2: Full-Rank and Curved Exponential Families

Assume $\mathcal{P} = \{P_\eta : \eta \in \Xi\}$ has densities

$$p_\eta(x) = e^{\eta' T(x) - A(\eta)} h(x)$$

with respect to some μ. If $T(x)$ satisfies no linear constraints; that is there do not exist an $a \neq 0, b$ with $a'T(x) \overset{\text{a.s.}}{=} b$, and if Ξ contains an open set, then we say that $\mathcal{P}$ is *full-rank*. Otherwise, we say that $\mathcal{P}$ is *curved*.

Note that this linear constraint condition just asks us to reduce the dimension of $\mathcal{P}$, so we can take WLOG $\mathcal{P}$ has this condition that $T(x)$ satisfies no linear constraints.

Theorem 8.3: Full Rank Families have Complete Sufficient Statistics

If $\mathcal{P}$ is full rank, then $T(X)$ is complete sufficient. That is, it is a complete sufficient statistic.

Proof: WLOG, assume that $\mathcal{P}$ is in canonical form (we can always reduce to this case)

$$p_\eta(x) = e^{\eta' x - A(\eta)}$$

Also, assume wlog that $0 \in \Xi^\circ$ and that $A(0) = 0$ (because we can always absorb any value of $A(0)$ into the carrier).

Now, suppose by way of contradiction that

$$\mathbb{P}_0(f(X) \neq 0) > 0$$

Then, this implies that $\mathbb{P}_\eta(f(X) \neq 0) > 0$ for all η, as our exponential tilt just scales this value. Also, assume that

$$\mathbb{E}_\eta[f(X)] = 0 \quad \forall \eta \in \Xi$$

Now, write $f(x) = f^+(x) - f^-(x)$ where $f^+ = f(x) \vee 0$ and $f^- = (-f(x)) \vee 0$. Then, $f^+, f^- \geq 0$. Then,

$$\int e^{\eta' x} f^+(x) d\mu(x) = \int e^{\eta' x} f^-(x) d\mu(x)$$

for all η. Now, let $Y^+ \sim f^+$ and $Y^- \sim f^-$. Now, as these have the same MGF in a neighborhood of 0 (by above), then $Y^+ \overset{D}{=} Y^-$ which implies that $f^+ \overset{\text{a.s.}}{=} f^-$. This implies that $f(x) \overset{\text{a.s.}}{=} 0$.

■

Theorem 8.4: Complete Sufficient implies Minimal Sufficient

If $T(X)$ is complete sufficient, then it is minimal sufficient.

Proof: Assume that $S(X)$ is minimal sufficient.

Let $\overline{T}(S(X)) = \mathbb{E}[T(X) \mid S(X)]$. We want to show that $\overline{T}(S(X)) \overset{\text{a.s.}}{=} T(X)$. Now, as S is minimal sufficient, we know that $S(X) \overset{\text{a.s.}}{=} f(T(X))$ for some f. Let

$$g(t) = t - \overline{T}(f(t))$$

Then,

$$\mathbb{E}_\theta[g(T(X))] = \mathbb{E}_\theta[T(X)] - \mathbb{E}_\theta[\overline{T}(S(X))]$$

However, as

$$\mathbb{E}_\theta[\overline{T}(S(X))] = \mathbb{E}_\theta[\mathbb{E}[T(X) \mid S(X)]] = \mathbb{E}_\theta[T(X)]$$

this implies that

$$\mathbb{E}_\theta[g(T(X))] = 0$$

which implies that $g(t) = 0$, so $T(X) \overset{\text{a.s.}}{=} \overline{T}(S(X))$.

Thus, this implies that T is minimal, as it is "isomorphic" to a minimal sufficient statistic.

■

Defintion 8.5: Ancillary Statistics

We say that $V(X)$ is *ancillary* for $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ if its distribution does not depend on θ.

What this implies is that $V(X)$ is a latent variable not depending on θ, so the conditional principle tells us to just condition on $V(X)$.

Theorem 8.6: Basu's Theorem

If $T(X)$ is complete sufficient for $\mathcal{P}$, and $V(X)$ is ancillary for $\mathcal{P}$, then $V \perp\!\!\!\perp T$ for all $P_\theta \in \mathcal{P}$.

Proof: What we want to show is that

$$\mathbb{P}_\theta(V \in A, T \in B) = \mathbb{P}_\theta(V \in A)\mathbb{P}_\theta(T \in B)$$

Define $q_A(T(x)) = \mathbb{P}(V \in A \mid T)$, and define $p_A = \mathbb{P}(V \in A)$. Notice that neither of these depend on θ, as T is sufficient, and V is ancillary. Now, their expectation's are equal, they are both equal to p_A (expectation of a condition is just expectation), which implies that $q_A(t) - p_A = 0$, which implies that

$$\mathbb{P}(V \in A \mid T) \overset{\text{a.s.}}{=} \mathbb{P}(V \in A)$$

■

Notice that the hypotheses of Basu's theorem is a statistics statement, whereas the conclusions tell us about probability. Thus, what this means is that if $\mathcal{P}$ doesn't satisfy the premises of Basu's theorem, we can rewrite

it as a union of families that do satisfy the premises of Basu's theorem, and we can get a weaker statement.

Example

Suppose that $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ where both $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ are unknown. Now, let

$$\overline{X} = \frac{1}{n}\sum X_i \quad \text{and} \quad S^2 = \frac{1}{n-1}\sum (X_i - \overline{X})^2$$

Define $\mathcal{P}_{\sigma^2} = \{P_{\mu,\sigma^2} : \mu \in \mathbb{R}\}$. That is, we restrict to a known σ^2. Then, it can be shown that $\overline{X}$ is complete sufficient by a theorem above, and S^2 is ancillary as it is independent for all $P_{\mu,\sigma^2} \in \mathcal{P}_{\sigma^2}$. Thus, they are independent, and as they are independent for all σ^2, they are independent everywhere.

■

9/26/26 Lecture 9: Rao-Blackwellization

Recall that with estimation, we will be using $\delta(X)$, an estimate (δ is called the estimator), to guess $g(\theta)$, an estimand. To test how good a given estimate is, we compute the loss $L(\theta, \delta(X))$. Then, to test how good an estimator is, we compute the risk:

$$R(\theta, \delta) = \mathbb{E}_\theta[L(\theta, \delta(X))]$$

There is no δ that minimizes risk everywhere, so we need to find a way to judge which estimators are better than others.

One way to do it is to summarize the risk with a value, whether it be the mean risk, or maximum possible risk.

Another way to do it is by restricting to a smaller class of estimators. We will first see what happens when we restrict to unbiased estimators. Recall that the bias of an estimator, $\text{Bias}(\theta, \delta)$ is defined to be

$$\text{Bias}(\theta, \delta) = \mathbb{E}_\theta[\delta(X)] - g(\theta)$$

we say an estimator is *unbiased* if $\text{Bias}(\theta, \delta) = 0$, so the average value of $\delta(X)$ is always $g(\theta)$.

Convex Loss

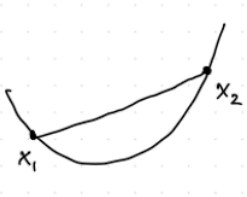
Definition 9.1: Convex Functions

A function $f(x)$ is *convex* if, for all x_1, x_2 and $\gamma \in (0, 1)$, we have that

$$f(\gamma x_1 + (1-\gamma)x_2) \leq \gamma f(x_1) + (1-\gamma) f(x_2)$$

The function f is *strictly convex* if equality never holds.

Here is what a convex (and in this case, strictly convex) function looks like visually. These secant lines always lie above the function.



Theorem 9.2: Jensen's Inequality

If f is convex, then $f(\mathbb{E}(X)) \leq \mathbb{E}f(X)$. If f is strictly convex, then $f(\mathbb{E}(X)) < \mathbb{E}f(X)$ unless X is almost surely constant.

This is useful since if we have a convex loss function, then we can find a lower bound on the risk by just taking the expectation of X and then plugging into the loss.

Note: We say that $L(\theta, d)$ is a convex loss function if it is convex in d (as we typically fix θ).

Example

Consider the squared error loss $L(\theta, d) = (g(\theta) - d)^2$, and the mean square error $\text{MSE}(\theta; \delta) = \mathbb{E}_\theta[(g(\theta) - \delta(X))^2]$. Recall that we can write

$$\text{MSE}(\theta; \delta) = \text{Bias}(\theta, \delta)^2 + \text{Var}_\theta(\delta(X))$$

Now, if δ is unbiased, then

$$\text{MSE}(\theta; \delta) = \text{Var}_\theta(\delta(X))$$

so the best unbiased estimator is the one that gives the least variance.

Theorem 9.3: Rao-Blackwell Theorem

Assume that $T(X)$ is sufficient, and $\delta(X)$ is any estimator. Let

$$\overline{\delta}(T(X)) = \mathbb{E}[\delta(X) \mid T(X)]$$

If $L(\theta, d)$ is convex, then $R(\theta, \overline{\delta}) \leq R(\theta, \delta)$ for all θ. Then, if $L(\theta, d)$ is strictly convex, then $R(\theta, \delta) < R(\theta, \delta)$ unless $\delta(X) \overset{\text{a.s.}}{=} \overline{\delta}(T(X))$ for all θ. We call $\overline{\delta}(T(x))$ is called the *Rao-Blackwellization* of $\delta(x)$.

Definition 9.4: U-estimability

An estimand $g(\theta)$ is *U-estimable* if there is some estimator $\delta(X)$ for which $\text{Bias}_\theta(\theta, \delta) = 0$ for all θ

Example

Let $X \sim \text{Bern}(\theta)$, $\theta \in (0, 1)$. Now, $\mathbb{E}_\theta[\delta(X)] = (1 - \theta)\delta(0) + \theta\delta(1)$, so estimators are just maps $\{0, 1\} \to \mathbb{R}$, and there is no way to have this be zero for all θ, so θ is not U-estimable.

Definition 9.5: Uniform Minimum Variance Unbiased (UMVU)

We say that $\delta(X)$ is *uniform minimum variance unbiased (UMVU)* if for any unbiased $\tilde{\delta}(X)$

$$\text{Var}_\theta(\delta(X)) \leq \text{Var}_\theta(\tilde{\delta}(X))$$

for all θ.

Theorem 9.6: Uniqueness of Rao-Blackwellization

Assume that $T(X)$ is complete sufficient, $g(\theta)$ is U-estimable and $L(\theta, d)$ is convex. Then, there is a unique unbiased estimator δ^* of the form $\delta^*(T(X))$ that uniformly minimizes risk (over all θ) among all unbiased estimators.

Proof Sketch: Since there is some unbiased estimator as $g(\theta)$ is U-estimable, then we can Rao-Blackwellize it. Then, it must be unique as $T(X)$ is complete. Then, it uniformly minimizes risk over all unbiased estimators because the Rao-Blackwellization of all unbiased estimators δ is δ^*, and this has less risk than δ^*.

Notice that Theorem 9.6 implies that δ^* also is UMVU as it minimizes the MSE and is unbiased.

Now, let's go into some examples to show the general process for finding the UMVU estimator. We will consider the case when $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Pois}(\theta)$. Notice that $\overline{X}$ is the UMVU for $g(\theta) = \theta$ as it is an unbiased estimator which is a function of the complete sufficient statistic (recall that $T(X) = \sum x_i$ is the complete sufficient statistic)

Now, what about estimators for $g(\theta) = \theta^2$? Is it naively $\overline{X}^2$? Well, let's use our methods to find it. Recall that the probability mass function as a function of $T(X)$ is

$$p_\theta(t) = \frac{e^{-n\theta}(n\theta)^t}{t!}$$

We will find the UMVU in two ways: Solving for an unbiased estimator based on T, and Rao-Blackwellizing any unbiased estimator.

Strategy 1: Create an Estimator as a function of T

Now, suppose that we have an unbiased estimator $\delta(T)$ as a function of the complete sufficient statistic. Then, we know that $\delta(T)$ is unbiased if, and only if

$$\begin{aligned}\theta^2 &= \int_{\{0,1,\dots\}} \delta(t) dP_\theta(t) \\ &= \sum_{t=0}^{\infty} \delta(t) p_\theta(t) \\ &= e^{-n\theta} \sum_{t=0}^{\infty} \delta(t) \frac{n^t}{t!} \theta^t\end{aligned}$$

Thus, we see that $e^{n\theta}\theta^2 = \sum_{t=0}^{\infty} \delta(t) \frac{n^t}{t!} \theta^t$. Then, we can expand the left hand side of this equation to get two power series

$$\sum_{k=0}^{\infty} \frac{n^k \theta^{k+2}}{k!} = \sum_{t=0}^{\infty} \delta(t) \frac{n^t \theta^t}{t!}$$

Now, change the index so the powers of θ start at the first term, which gives

$$\sum_{t=2}^{\infty} \frac{n^{t-2}\theta^t}{(t-2)!} = \sum_{t=0}^{\infty} \delta(t) \frac{n^t}{t!} \theta^t$$

Matching terms, we must have that $\delta(0) = 0$, $\delta(1) = 0$, and for $t \geq 2$, we must have

$$\delta(t) = \frac{t(t-1)}{n^2}$$

Thus, that is our estimator!

$$\delta(T(X)) = \frac{T(X)(T(X)-1)}{n^2} = \overline{X}^2 - \frac{\overline{X}}{n}$$

since $\overline{X} = \frac{T(X)}{n}$.

Strategy 2: Rao-Blackwellize an Unbiased Estimator

Let $\delta(X) = X_1 \cdot X_2$. Since $\mathbb{E}[\delta(X)] = \theta^2$, this is an unbiased estimator (although a bad one). Now, let's Rao-Blackwellize. Let

$$\delta^*(T) = \mathbb{E}[X_1 X_2 \mid T]$$

Recall that $X \mid T \sim \text{Multinom}(t, \frac{1}{n}1_n)$ where 1_n is a n-vector of 1's. Thus, we see that $X_1 \mid T \sim \text{Binom}(t, \frac{1}{n})$. Thus, we see that

$$\mathbb{E}[X_1 \mid T] = \frac{T}{n} \quad \text{and} \quad \text{Var}(X_1 \mid T) = T \cdot \frac{1}{n} \cdot \left(1 - \frac{1}{n}\right) = \frac{T(n-1)}{n^2}$$

Then, as $X_2 \mid T, X_1 \sim \text{Binom}(t - x_1, \frac{1}{n-1})$, we have that

$$\mathbb{E}[X_2 \mid T, X_1] = \frac{T - X_1}{n-1}$$

Thus, we have that

$$\begin{aligned}
\mathbb{E}[X_1 X_2 \mid T] &= \mathbb{E}[X_1 \cdot \mathbb{E}[X_2 \mid T, X_1] \mid T] \\
&= \mathbb{E}\left[X_1 \cdot \frac{T - X_1}{n-1} \mid T\right] \\
&= \mathbb{E}\left[\frac{TX_1}{n-1} - \frac{X_1^2}{x-1} \mid T\right] \\
&= \frac{T(T-1)}{n^2}
\end{aligned}$$

which is the same thing we got before!

Trick Strategy 3: Fun trick

There is an even easier way to do this (just by coincidence that $\overline{X}^2$ is a moment). Notice that as

$$\mathbb{E}[\overline{X}^2] = \text{Var}(\overline{X}) + (\mathbb{E}[\overline{X}])^2 = \frac{\theta}{n} + \theta^2$$

we see that

$$\mathbb{E}\left[\overline{X}^2 - \frac{\overline{X}}{n}\right] = \theta^2$$

so as this is the expectation of $\delta(T) = \frac{T^2}{n^2} - \frac{T}{n^2}$, we see it is unbiased and as it is a function of T, is is the UMVU.

Here is one last interesting example.

Example

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} U[0, \theta]$ with $\theta > 0$. Now, recall that $T = X_{(n)}$ is complete sufficient with

$$p_\theta^T(X) = \frac{n}{\theta^n} t^{n-1} 1\{t \leq \theta\}$$

Then, notice we have that

$$\begin{aligned}
\mathbb{E}_\theta[T] &= \int_0^\theta t p_\theta(t) dt \\
&= \int_0^\theta \frac{n}{\theta^n} t^n dt \\
&= \left[\frac{n t^{n+1}}{(n+1)\theta^n}\right]_0^\theta \\
&= \frac{n}{n+1}\theta
\end{aligned}$$

Thus, we then see that

$$\mathbb{E}_\theta\left[\frac{n+1}{n}T\right] = \theta$$

so $\delta(T) = \frac{n+1}{n}T$ must be the UMVU as it is an unbiased estimator of T.

■

9/29/26 Lecture 10: Score and Fisher Information

Recall last lectures example: $X_1, \dots, X_n \overset{\text{iid}}{\sim} U[0,1]$. Recall that $X_{(n)}$ is complete sufficient, and as

$$\mathbb{E}_\theta[X_{(n)}] = \frac{n}{n+1}\theta \implies \frac{n+1}{n}X_{(n)} \text{ is UMVU}$$

Now, also notice that, if $Z = \frac{X_{(n)}}{\theta} \sim \text{Beta}(n,1)$, then

$$\begin{aligned}\text{MSE}(\theta; cX_{(n)}) &= \mathbb{E}_\theta[(cX_{(n)} - \theta)^2] \\ &= \theta^2\mathbb{E}[(cZ-1)^2] \\ &= \theta^2 \text{MSE}(1; cZ)\end{aligned}$$

Where the latter is something independent on θ. Thus, there is some optimal c. Well, we see that

$$\text{MSE}(1; cZ) = c^2\mathbb{E}[Z^2] - 2c\mathbb{E}[Z] + 1$$

So minimizing c, we get the best possible c is $c = \frac{n+2}{n+1}$. Notice that we don't get the same c as above, namely $\frac{n}{n+1}$, so our UMVU above doesn't not minimize MSE across all estimators, but also notice that $\frac{n+2}{n+1}X_{(n)}$ is unbiased. Thus, going towards unbiasedness does not necessarily mean going towards minimizing MSE.

Here is another example where restricting to unbiased estimators is not a good idea. Let $X \sim N_d(\mu_i, I_d)$ for $\mu \in \mathbb{R}^d$ and suppose we want to estimate $g(\mu) = \|\mu\|^2$. It turns out that X is a complete sufficient statistic. Now, we see that, defining $Z = N(0, I_d)$,

$$\begin{aligned}\mathbb{E}_\mu\|X\|^2 &= \sum_{j=1}^d \mathbb{E}[(\mu_j + Z_j)^2] \\ &= \sum_{j=1}^d \mu_j^2 + 1 = \|\mu\|^2 + d\end{aligned}$$

Thus, our estimator should be $\delta(X) = \|X\|^2 - d$, which is the best unbiased, but this is a bad idea as we might get some negative observations. Thus, we see that $\delta^*(X) = (\|X\|^2 - d)_+$ is better for all μ and all X.

Finally, here's one more absurd example. Let $X \sim \text{Binom}(1000, \theta)$, and we want to estimate $g(\theta) = \mathbb{P}_\theta(X \geq 500)$. Consider the following estimator

$$\delta(X) = 1\{X \geq 500\}$$

This is unbiased and is a function of the complete sufficient statistic (which is X), but this is a crazy estimator! It means that if we observe 499 heads, then we estimate that $\mathbb{P}_\theta(X \geq 500) = 0$, and if we observe 500 heads, then we estimate that $\mathbb{P}_\theta(X \geq 500) = 1$. That's insane!

The idea behind why these all suck is that it's too easy to be a UMVU. All you need is that your estimator is a function of the complete sufficient statistic and is unbiased, which can lead to easily constructing many examples where this sucks.

Score Function

Assume $\mathcal{P}$ has densities p_θ with respect to μ, and $\Theta \subseteq \mathbb{R}^d$, and assume that p_θ hsa a common support for all θ. Now, we have

$$\ell(\theta, X) = \log p_\theta(X)$$

is a random function of θ. Then, notice that $\ell(\cdot, X) - \ell(\theta_0, X)$ is minimal sufficient for any fixed reference value $\theta_0 \in \Theta$.

In the asymptotic picture; that is when we have a lot of data and we can narrow the possible θ down to a small neighborhood of the true value, and if θ_0 is the true value, the "effective model" is $\{P_{\theta_0+\eta} : \eta \approx 0\}$. Then, via a Taylor approximation,

$$\ell(\theta_0 + \eta, X) - \ell(\theta_0, X) \approx \eta' \nabla_\theta \ell(\theta_0, X) + O(\|\eta\|^2)$$

Since we know the LHS is minimal sufficient, $\nabla_\theta \ell(\theta_0, X)$ should be close to being minimal sufficient locally.

Definition 10.1: Score

The *score* at θ_0 is

$$S_{\theta_0}(X) = \nabla_\theta \ell(\theta_0, X)$$

Now, note from the score we also have

$$p_{\theta_0+\eta}(x) = e^{\ell(\theta_0+\eta, x)} \approx e^{\eta' S_{\theta_0}(x) + O(\|\eta\|^2)} p_{\theta_0}(x)$$

Thus, we see that the score locally transforms p_θ into an exponential family!

Differential Identities

Now, assuming that everything is regular and extends smoothly, we see that

$$\begin{aligned} 1 &= \int_{\mathcal{X}} e^{\ell(\theta, X)} d\mu(x) \\ 0 &= \int_{\mathcal{X}} \frac{\partial}{\partial \theta_i} \ell(\theta, x) e^{\ell(\theta, x)} d\mu(x) && \text{(differentiate wrt } \theta_i\text{)} \end{aligned}$$

Thus, collecting this into a vector, we have that

$$0 = \mathbb{E}_\theta[\nabla \ell(\theta, X)] = \mathbb{E}_\theta[S_\theta(X)]$$

Then, taking another derivative, we have

$$\begin{aligned} 0 &= \int_{\mathcal{X}} \left(\frac{\partial^2 \ell}{\partial \theta_j \partial \theta_k} + \frac{\partial \ell}{\partial \theta_j} \frac{\partial \ell}{\partial \theta_k} \right) e^{\ell(\theta, x)} d\mu(x) \\ &= \mathbb{E}_\theta \left[\frac{\partial^2 \ell}{\partial \theta_i \partial \theta_k}(\theta, X) \right] + \mathbb{E}_\theta \left[\frac{\partial \ell}{\partial \theta_j}(\theta, X) \frac{\partial \ell}{\partial \theta_k}(\theta, X) \right] \\ &= \mathbb{E}_\theta \left[\frac{\partial^2 \ell}{\partial \theta_i \partial \theta_k}(\theta, X) \right] + \operatorname{Cov}\left(\frac{\partial \ell}{\partial \theta_i}, \frac{\partial \ell}{\partial \theta_k} \right) && \text{(as the expectation of single derivatives is 0)} \end{aligned}$$

Thus, we see that

$$\operatorname{Var}_\theta\left(\nabla \ell(\theta, X)\right) = \mathbb{E}_\theta\left[-\nabla^2 \ell(\theta, X)\right]$$

Definition 10.2: Fisher Information

We call $J(\theta) = \mathrm{Var}_\theta(S_\theta(X))$ the *Fisher information.*

Here's an example

Example

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} p_\theta^{(i)}(x)$ for $\theta \in \Theta \subseteq \mathbb{R}^d$. Then, suppose that p_θ has nice regular qualities, like common supports and that $\mathbb{E}_\theta \|\nabla_\theta \ell(\theta, X)\|^2 < \infty$. Then, we see that

$$X \sim p_\theta(X) = \prod p_\theta^{(i)}(x_i)$$

Then if

$$\ell_i(\theta_i, x_i) := \log p_\theta^{(i)}(x_i)$$

we see that

$$\ell(\theta, x) = \sum_i \ell_i(\theta_i, x_i)$$

Thus, we have that

$$S_\theta(x) = \nabla \ell(\theta, X) = \sum_i S_\theta^{(i)}(x_i)$$

Additionally, we have that

$$\begin{aligned} J(\theta) &= \mathrm{Var}_\theta S_\theta(X) \\ &= n \mathrm{Var}_\theta S_\theta^{(i)}(X_i) \\ &= n J_1(\theta) \end{aligned}$$

by the fact the X_i are iid.

Now, in "asymptopia", we see that

$$\frac{1}{n} \nabla^2 \ell(\theta, X) = \frac{1}{n} \sum_{i=1}^n \nabla^2 \ell(\theta_i, X_i) \to -J_1(\theta)$$

Thus, we see that

$$O(\|\eta\|^2) = \frac{1}{2} \eta' (n J_1(\theta)) \eta'$$

since we have, by a Taylor expansion,

$$\begin{aligned} \ell\left(\theta_0 + \frac{\eta}{\sqrt{n}}, X\right) - \ell(\theta_0, X) &\approx \frac{\eta}{\sqrt{n}}' S_{\theta_0} S(X) + \frac{1}{2} \eta' \left(\frac{\sum \nabla^2 \ell(\theta_i, X_i)}{n} \right) \eta \\ &\approx \eta' \frac{S_{\theta_0}(X)}{\sqrt{n}} - \frac{1}{2} \eta' J_1(\theta) \eta \\ &\approx \eta' Z_{\theta_0}(X) - \frac{1}{2} \eta' J_1(\theta) \eta \end{aligned}$$

Then as

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n S_{\theta_0}^{(i)}(X_i) \approx N(0, J_1(\theta))$$

Thus, looking back at our representation as an exponential family, we see that the minimal sufficient statistic is

$$Z_{\theta_0}(X) \sim N(0, J_1(\theta))$$

10/1/26 Lecture 11: Cramer-Rao Lower Bound

One minimal sufficient statistic is $\ell(\theta; x)$, but we can also consider $\ell(\theta; x) - \ell(\theta_0, x)$, which is nice as it forces this statistic to be zero at θ_0.

Recall the score and fisher information:

$$S_\theta(x) = \nabla \ell(\theta, x)$$
$$\mathbb{E}_\theta S_\theta(X) = 0$$

and

$$J(\theta) = \mathrm{Var}_\theta(S_\theta(X)) = -\mathbb{E}_\theta \nabla^2 \ell(\theta, X)$$

Given an IID sample, we see that

$$S_\theta(X) = \sum_{i=1}^n S_\theta^{(i)}(X_i)$$

and

$$J(\theta) = nJ_1(\theta)$$

Now, if we reparameterize and let $\theta(\eta) = \theta_0 + \frac{\eta}{\sqrt{n}}$, then we see that

$$\begin{aligned}
\ell(\theta(\eta)) - \ell(\theta_0) &= (\theta - \theta_0) S_\theta(x) + \frac{1}{2}(\theta - \theta_0)^2 \nabla^2 \ell(\theta_0, X) \\
&= \eta' \frac{1}{\sqrt{n}} \sum S_\theta^{(i)}(X_i) - \frac{1}{2}\eta' \frac{1}{n} \left(nJ_1(\theta) + \sum (\nabla^2 \ell(\theta, X) - J_1(\theta)) \right) \eta \\
&= \eta' Z - \frac{1}{2}\eta' J_1(\theta)\eta + \frac{1}{2}\eta' \left(\frac{1}{n} \sum_{i=1}^n \epsilon_i \right) \eta
\end{aligned}$$

Taking n to infinity, the error term dissapears and we see that heuristically, the sufficient statistic approximately $S_\theta(x)/\sqrt{n}$. Exponentiating, and moving things around, we see that

$$p_{\theta_0 + \eta/\sqrt{n}}(x) \approx e^{\eta' \frac{S_\theta(x)}{\sqrt{n}} - \frac{1}{2}\eta' J_1(\theta)\eta} p_{\theta_0}(x)$$

so can approximate p_θ with an exponential family (at least locally).

Cramer-Rao Lower Bound

Suppose that $\delta(X)$ is unbiased for $g(\theta) \in \mathbb{R}$. Then

$$g(\theta) = \int \delta(x) e^{\ell(\theta, x) d\mu(x)}$$

Taking derivatives,

$$\frac{\partial g}{\partial \theta_j}(\theta) = \int \delta(X) \frac{\partial \ell}{\partial \theta_j}(\theta) e^{\ell(\theta, x)} d\mu(x) = \mathbb{E}_\theta[\delta(X) \cdot \frac{\partial \ell}{\partial \theta_j}(\theta, x)] = \mathrm{Cov}_\theta \left(\delta(X), \frac{\partial \ell}{\partial \theta_i} \right)(\theta, X)$$

Thus,

$$\nabla g(\theta) = \mathbb{E}_\theta[\delta(X) S_\theta(X)] = \mathrm{Cov}_\theta(\delta(X), S_\theta(x))$$

Now, we know that, in 1 dimension

$$1 \geq \mathrm{Corr}_\theta(\delta(X), S_\theta(X)) = \frac{\mathrm{Cov}_\theta^2(\delta, S_\theta)}{\mathrm{Var}_\theta(\delta) \cdot \mathrm{Var}_\theta(S_\theta)} = \frac{(\nabla g(\theta))^2}{\mathrm{Var}_\theta(\delta) J(\theta)}$$

Thus, we see that

$$\text{Var}_\theta(\delta(X)) = \frac{(\dot{g}(\theta))^2}{J(\theta)\text{Corr}_\theta(\delta(X), S_\theta(X))} \geq \frac{(\dot{g}(\theta))^2}{J(\theta)}$$

If $g(\theta) = \theta$ and $X = (X_1, \ldots, X_n)$ an iid sample, we see that

$$Var_\theta(\delta(X)) \geq \frac{1}{nJ_1(\theta)} \implies \text{SD}(\delta(X)) \approx \frac{1}{\sqrt{n}}$$

and if $\theta \in \mathbb{R}^d$ and $g(\theta) \in \mathbb{R}$,

$$\text{Var}_\theta(\delta(X)) \geq \nabla g(\theta)' J(\theta)^{-1} \nabla g(\theta)$$

and the reason for this is that for $a \in \mathbb{R}^d$,

$$\text{Cov}_\theta(\delta, a'S_\theta) = \mathbb{E}_\theta[\delta(a'S_\theta)] = a'\mathbb{E}_\theta[\delta S_\theta] = a'\nabla g(\theta)$$

so by the 1-dimensional result above, we have that

$$\begin{aligned}\text{Var}_\theta(\delta(X)) &\geq \frac{\text{Cov}_\theta(\delta, a'S_\theta)^2}{\text{Var}_\theta(a'S_\theta)} \\ &= \frac{a'\nabla g(\theta)\nabla g(\theta)'a}{a'\text{Var}_\theta(S_\theta)a}\end{aligned}$$

Maximizing over a, we get the maximal a to be $a^* = J^{-1}(\theta)\nabla g(\theta)$ so plugging in a^*, we get the desired result. Now, let's consider the exponential family. Let

$$p_\eta(x) = e^{\eta' T(x) - A(\eta)} h(x)$$

then

$$\ell(\eta, X) = \eta' T(x) - A(\eta) + \log h(x)$$

Taking derivatives with respect to η,

$$\nabla \ell(\eta, X) = T(X) - \nabla A(\eta) = T(X) - \mathbb{E}_\eta(T(X))$$

Thus, the score is playing the role of $T(X)$ (up to forcing unbiasedness), the same as we saw before. Similarly, we see that

$$\text{Var}_\eta(S_\eta(X)) = \text{Var}_\eta(T(X)) = \nabla^2 A(\eta)$$

and

$$\nabla^2 \ell(\eta, X) = -\nabla^2 A(\eta)$$

Thus, $J(\eta) = -\nabla^2 \ell(\eta, X)$, which was as expected. Now, we can also represent this family as

$$e^{(\eta/c)'(cT(x)) - A(\eta/c)}$$

we see that $\nabla^2 \ell(\eta, X) = -c^2 \nabla(A\eta) = c^2 \text{Var}_{c\eta}(T(X))$. This seems odd, but notice that we are working in "different units". Heuristically, when we parameterize like this, changing the new parameter will change the distribution differently, and this factor accounts for that.

Example

Consider

$$p_\theta(x) = e^{\eta(\theta)' T(x) - A(\eta(\theta))} h(x)$$

and consider $\theta \in \mathbb{R}$ and $\eta(\theta) \in \mathbb{R}^d$ (a curved family). Then,

$$\ell(\theta, X) = \eta(\theta)' T(X) - A(\eta(\theta)) + \log h(x)$$

so

$$\dot{\ell}(\theta, X) = \dot{\eta}(\theta)'T(X) - \dot{\eta}(\theta)'\nabla A(\eta(\theta)) = \dot{\eta}(\theta)'(T - \mathbb{E}_\theta T)$$

Then, we see that

$$\dot{\eta}(\theta)'(T - \mathbb{E}_\theta(T)) = \dot{\eta}(\theta)' S^{(\eta)}_{\eta(\theta)}(X)$$

and we see that

$$\ddot{\ell}(\theta, X) = \ddot{\eta}(\theta)'(T(X) - \mathbb{E}_\theta(T(X))) - \dot{\eta}(\theta)'\nabla^2 A(\eta(\theta))\dot{\eta}(\theta)$$

so

$$\mathbb{E}_\theta[-\ddot{\ell}(\theta)] = \dot{\eta}(\theta)'\nabla^2 A(\eta(\theta))\dot{\eta}(\theta) = \dot{\eta}(\theta)' J^{(\eta)}(\eta(\theta))\dot{\eta}(\theta)$$

Bayes Estimators

Consider $X \sim P_\theta$ and $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$, with a loss $L(\theta, d)$ and $R(\theta, \delta()) = \mathbb{E}_\theta[L(\theta, \delta(X))]$.

Definition 11.1: Bayes Risk

We define the *Bayes risk*

$$r_\Lambda(\delta) = \int_\Theta R(\theta, \delta) d\Lambda(\theta)$$

where the measure Λ is called the *prior*. If $\Lambda(\Theta) = \infty$, we call this prior *improper*, and if $\Lambda(\Theta) < \infty$, we can take WLOG it to be 1, and we call this prior *proper*.

Notice that

$$r_\Lambda(\delta) = \mathbb{E}_{\theta\sim\Lambda}[R(\theta; \delta)] = \mathbb{E}[L(\theta, \delta(X))]$$

assuming that $\theta \sim \Lambda$ and $X \mid \theta \sim P_\theta$. This seems very Bayesian, but this is just mathematical manipulations. Of course, if you were Bayesian, this tells you exactly how good an estimator is, but if you're frequentist, this is just putting an arbitrary distribution and testing δ.

Definition 11.2: Posterior

The conditional distribution $\Lambda(\theta \mid X)$ is called the *posterior*.

Assuming that the prior has density $\lambda(\theta)$ and the likelihood is $p_\theta(X)$, then the joint density is

$$\lambda()$$

10/6/26 Lecture 12: More on Bayes Estimators

Example: Curved Gaussian Family

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} N_d(\mu(\theta), I_d)$, where $\mu : \mathbb{R} \to \mathbb{R}^d$.

Then, in our ambient family,

$$S_\mu^{(\mu)}(X) = \sum X_i - n\mu = n(\overline{X} - \mu)$$

and

$$J^{(\mu)}(\mu) = nI_d$$

An in our curved family,

$$S_\theta(X) = \dot{\mu(\theta)}'(\overline{X} - \mu(\theta))$$

and

$$J(\theta) = n\|\dot{\mu}(\theta)\|^2$$

Now, suppose that $\mu(\theta) = (\cos\theta \ \sin\theta)$, for $\theta \in [0, \pi/2]$. For $\theta = 0$, we see that $\mu(0) = (1\ 0)$ and $\dot{\mu}(0) = (0\ 1)$ so by a quick computation $S_0(X) = n\overline{X_2}$. Similarly, for $\theta = \pi/2$, then $\mu(\pi/2) = (0\ 1)$ and $\dot{\mu}(\pi/2) = (-1\ 0)$, so $S_{\pi/2}(X) = -n\overline{X_1}$.

Going back to Bayes risk, recall we have the Bayes error

$$r_\Lambda(\delta) = \int R(\theta; \delta(\cdot))d\Lambda(\theta) = \mathbb{E}[L(\theta, \delta(X))]$$

where this expectation is with respect to $\theta \sim \Lambda$ and $X \mid \theta \sim P_\theta$.

We also have our prior $\lambda(\theta)$, likelihood $p_\theta(x)$ and marginal $q(x) = \int \lambda(\theta)p_\theta(x)d\theta$. We also have our posterior

$$\lambda(\theta \mid X) = \lambda(\theta)\frac{p_\theta(X)}{q(X)}$$

What we will do is try to minimize $\mathbb{E}[L(\theta, \delta(X))]$ X by X and then piece this together.

Theorem 12.1: Existance of Bayes Estimator

Consider the model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ with a proper prior $\Lambda(\theta)$. Suppose that $L(\theta, d) \geq 0$ and that $r_\Lambda(\delta_0) < \infty$ for some δ_0. Then, it turns out that $\delta_\Lambda(x)$ is Bayes (bayes risk minimizing), with $r_\Lambda(\delta_\Lambda) < \infty$ if, and only if,

$$\delta_\Lambda(x) \in \operatorname*{argmin}_d \mathbb{E}[L(\theta, d) \mid X = x]$$

for almost every x. That, is $\delta_\Lambda(x)$ is Bayes if it is argmin almost everywhere.

Proof: ($\Rightarrow$) Suppose that $\delta(X)$ is some other estimator, so almost everywhere

$$\mathbb{E}[L(\theta, \delta(x)) \mid X = x] \geq \mathbb{E}[L(\theta, \delta_\Lambda(x)) \mid X = x]$$

Then, taking the expectation over X, we see that $r_\Lambda(\delta) \geq r_\Lambda(\delta_\Lambda)$, so δ_Λ is Bayes risk minimizing, which implies it is a Bayes estimator.

($\Leftarrow$) Assume that $\delta_\Lambda(x) < \infty$ is a Bayes estimator (as otherwise, δ_0 would be better). Then, defining

$$E_x(d) := \mathbb{E}[L(\theta, d) \mid X = x]$$

Now, let

$$A_\epsilon = \{x : E_x(\delta_\Lambda(x) > \epsilon + \inf_d E_x(d))\}$$

Now, suppose there exists an $\epsilon > 0$ such that $\mathbb{P}(X \in A_\epsilon) > 0$. Then, on A_ϵ, we can choose a $\delta^*(X)$ such that

$$\delta^*(X) \leq E_x(\delta_\Lambda(x)) - \epsilon$$

by the definition of an inf. Then, on A_ϵ^c, take $\delta^*(x) = \delta_\Lambda(x)$. Then, we see that

$$r_\Lambda(\delta_\Lambda) - r_\Lambda(\delta^*) \geq \mathbb{P}(X \in A)\epsilon$$

This contradicts that δ_Λ is a Bayes estimator. Thus, no epsilon exists, so A_ϵ must have measure zero for all ϵ.

If we choose our loss to be $L(\theta, d) = (g(\theta) - d)^2$, then

$$\delta_\Lambda(x) = \mathbb{E}[g(\theta) \mid X]$$

This is because

$$\begin{aligned}
\mathbb{E}[(g(\theta) - d)^2 \mid X] &= \mathbb{E}[(g(\theta) - \mathbb{E}[g(\theta) \mid X] + \mathbb{E}[g(\theta) \mid X] - d)^2 \mid X] \\
&= \mathrm{Var}(g(\theta) \mid X) + (\mathbb{E}[g(\theta) \mid X] - d)^2 + 2\mathbb{E}((g(\theta) - \mathbb{E}[g(\theta) \mid X])(\mathbb{E}[g(\theta) \mid X] - d) \mid X) \\
&= \mathrm{Var}(g(\theta) \mid X) + (\mathbb{E}[g(\theta) \mid X] - d)^2
\end{aligned}$$

Then, if we let $\delta_\Lambda(X) = \mathbb{E}[g(\theta \mid X)]$, this minimizes the integrand of $r_\Lambda(\delta)$ as the second term becomes zero, and we see that

$$r_\Lambda(\delta_\lambda) = \mathbb{E}[\mathrm{Var}(g(\theta) \mid X)]$$

Now, suppose that

$$L(\theta, d) = w(\theta)(g(\theta) - d)^2$$

which is reasonable as when $g(\theta)$ is really big, it might make sense to scale the errors down. For instance, we might want to look at

$$L(\theta, d) = \frac{\theta - d}{\theta}^2$$

Now, we see that

$$\mathbb{E}[(d - g(\theta))^2 w(\theta) \mid X] = d^2\mathbb{E}[w(\theta) \mid X] - 2d\mathbb{E}[w(\theta)g(\theta) \mid X] + \mathbb{E}[w(\theta)g(\theta)^2 \mid X]$$

Taking a derivative with respect to d to and setting this equal to zero, we find that the Bayes estimator must be

$$\delta_\Lambda(X) = \frac{\mathbb{E}[w(\theta)g(\theta) \mid X]}{\mathbb{E}[w(\theta) \mid X]} = \frac{\int g(\theta)w(\theta)\lambda(\theta \mid x)d\theta}{\int w(\theta)\lambda(\theta \mid x)d\theta}$$

Then, simplifying the RHS,

$$\begin{aligned}
\frac{\int g(\theta)w(\theta)\lambda(\theta \mid x)d\theta}{\int w(\theta)\lambda(\theta \mid x)d\theta} &= \frac{\int g(\theta)w(\theta)\lambda(\theta)p_\theta(x)d\theta}{\int w(\theta)\lambda(\theta)p_\theta(x)d\theta} \\
&= \frac{\int g(\theta)(\lambda(\theta)w(\theta))p_\theta(x)d\theta}{\int \lambda(\theta)w(\theta)p_\theta(x)d\theta}
\end{aligned}$$

where the last integral is $\mathbb{E}[g(\theta) \mid X]$ where θ is sampled from $(\lambda \cdot w)(\theta)$ and $X \mid \theta \sim p_\theta(x)$. Thus, scaling the loss is effectively changing the prior.

Example: Beta Binomial

Suppose $X \mid \theta \sim \text{Binom}(n, \theta)$ and $\theta \sim \text{Beta}(\alpha, \beta)$. Now, we have The first has density $\theta^x(1-\theta)^{n-x}\binom{n}{x}$ and the second has density $\theta^{\alpha-1}(1-\theta)^{\beta-1}\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$. We see that

$$\lambda(\theta \mid X) = \lambda(\theta)\frac{p_\theta(x)}{q(x)}$$

we will ignore $q(x)$ and other constants and recover it by renormalizing. Thus, we see that

$$\lambda(\theta \mid X) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}\theta^x(1-\theta)^{n-x} = \theta^{x+\alpha-1}(1-\theta)^{n-x+\beta-1}$$

Thus, this is proportional to $\text{Beta}(x+\alpha, n-x+\beta)$, which implies that once we normalize, we see that $\lambda(\theta \mid X)$ is $\text{Beta}(x+\alpha, n-x+\beta)$, and knowing this, we can compute that

$$\mathbb{E}[\theta \mid X] = \frac{X+\alpha}{n+\alpha+\beta}$$

Thus, from this result, we see that $\frac{X+1}{n+2}$ is the Bayes estimator for $\Lambda = U[0,1]$, and $\frac{X+2}{n+4}$ is the Bayes estaimtor for $\Lambda = \text{Beta}(2,2)$. Now, notice that we can write

$$\mathbb{E}[\theta \mid X] = \frac{X}{n} \cdot \frac{n}{n+\alpha+\beta} + \frac{\alpha}{\alpha+\beta} \cdot \frac{\alpha+\beta}{n+\alpha+\beta}$$

which is a convex combination of actual data (X/n) and pseudo-data $(\alpha/(\alpha+\beta))$.

10/8/26 Lecture 13: Conjugate Priors

Recall from last time that if $X \mid \theta \sim \text{Binom}(n, \theta)$ and $\theta \sim \text{Beta}(\alpha, \beta)$, then

$$\begin{aligned}\lambda(\theta \mid X) &\propto_\theta \lambda(\theta) p_\theta(X) \\ &\propto_\theta \text{Beta}(\alpha + X, \beta + n - X)\end{aligned}$$

which is very nice, as it gives us

$$\mathbb{E}[\theta \mid X] = \frac{X + \alpha}{n + \alpha + \beta}$$

Example: Normal Location

Let $X \mid \theta \sim N(\theta, \sigma^2)$ and $\theta \sim N(\mu, \tau^2)$, where we assume σ^2 and τ^2 are known. That is,

$$p_\theta(x) \propto e^{-(x-\theta)^2/2\sigma^2} \quad \text{and} \quad \lambda(\theta) \propto e^{-(\theta-\mu)^2/2\tau^2}$$

Then,

$$\begin{aligned}\lambda(\theta \mid x) &\propto_\theta \exp\left\{-\frac{(x-\theta)^2}{2\sigma^2} - \frac{(\theta-\mu)^2}{2\tau^2}\right\} \\ &\propto_\theta \exp\left\{\frac{x\theta}{\sigma^2} - \frac{\theta^2}{2\sigma^2} - \frac{\theta^2}{2\tau^2} + \frac{\theta\mu}{\tau^2}\right\} \\ &\propto_\theta \exp\left\{-\left(\theta - \frac{x\sigma^{-2} + \mu\tau^{-2}}{\sigma^{-2} + \tau^{-2}}\right)^2 \Big/ 2(\sigma^{-2} + \tau^{-2})^{-1}\right\} \\ &\propto_\theta N\left(\frac{x\sigma^{-2} + \mu\tau^{-2}}{\sigma^{-2} + \tau^{-2}}, \frac{1}{\sigma^{-2} + \tau^{-2}}\right)\end{aligned}$$

Notice that the posterior mean is a precision weighted average, where σ^{-2} is the likelihood precision and τ^{-2} is the prior precision. The posterior variance is the harmonic sum of σ^2 and τ^2, so $\sigma^{-2} + \tau^{-2}$ is the posterior precision. Notice that

$$\mathbb{E}[\theta \mid X] = X \cdot \frac{\sigma^{-2}}{\sigma^{-2} + \tau^{-2}} + \mu \cdot \frac{\tau^{-2}}{\sigma^{-2} + \tau^{-2}} \quad \text{and} \quad \text{Var}(\theta \mid X)^{-1} = \sigma^{-2} + \tau^{-2}$$

Now, if $\sigma^2 = 1$ and $k = \frac{1}{\tau^2}$, then

$$\mathbb{E}[\theta \mid X] = X \cdot \frac{1}{1+k} + \mu \cdot \frac{k}{1+k} \quad \text{and} \quad \text{Var}(\theta \mid X)^{-1} = 1 + k$$

so once again, we can think of μ as "pseudo-data".

Now, suppose $X_1, \ldots, X_n \mid \theta \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$ and $\theta \sim N(\mu, \tau^2)$. Since using a minimal sufficient statistic doesn't change the posterior, we can do a sufficiency reduction to

$$\overline{X} \mid \theta \sim N(\theta, \sigma^2/n), \quad \theta \sim N(\mu, \tau^2)$$

so by the above,

$$\mathbb{E}[\theta \mid X] = \overline{X} \cdot \frac{n\sigma^{-2}}{n\sigma^{-2} + \tau^{-2}} + \mu \cdot \frac{\tau^{-2}}{n\sigma^{-2} + \tau^{-2}} \quad \text{and} \quad \text{Var}(\theta \mid X)^{-1} = n\sigma^{-2} + \tau^{-2}$$

Again, if $\sigma^2 = 1$ and $k = \frac{1}{\tau^2}$, then

$$\mathbb{E}[\theta \mid X] = \overline{X} \cdot \frac{n}{n+k} + \mu \cdot \frac{k}{n+k} \quad \text{and} \quad \text{Var}(\theta \mid X)^{-1} = n + k$$

which confirms the belief that we are adding "psuedo data".

Remark

If $n \gg k$, the data "swamps" the prior, and if $k \gg n$, the prior "swamps" the data.

Remark

It seems odd that we always get lucky in the sense that our posterior has the same distribution of the prior. This is because of exponential families, as we will see next!

Conjugate Priors

Suppose $X_1, \dots, X_n \mid \eta \overset{\text{iid}}{\sim} p_\eta(x) = e^{\eta' T(x) - A(\eta)} h(x)$, with $\eta \in \Xi \subseteq \mathbb{R}^s$. For a carrier $\lambda_0(\eta)$, define the $(s+1)$-parameter exponential family by

$$\lambda_{\mu,k}(\eta) = e^{k\mu'\eta - kA(\eta) - B(k\mu,k)} \lambda_0(\eta)$$

This family has sufficient statistic $\begin{pmatrix} \eta \\ -A(\eta) \end{pmatrix} \in \mathbb{R}^{s+1}$ and natural parameter $\begin{pmatrix} k\mu \\ k \end{pmatrix}$.

Now, what is the posterior? We see that

$$\begin{aligned} \lambda(\eta \mid X_1, \dots, X_n) &\propto_\eta e^{k\mu'\eta - kA(\eta)} \lambda_0(\eta) \cdot \prod_{i=1}^n e^{\eta' T(x_i) - A(\eta)} h(x_i) \\ &\propto_\eta \lambda_{\mu^+, k^+}(\eta) \end{aligned}$$

where

$$\mu^+ = \frac{k\mu + \sum T(x_i)}{k+n}, \quad k^+ = k + n$$

Thus, the posterior stays in the family of the prior, and the posterior looks like what happens when you add $k\mu$ as pseudo data.

Definition 13.1: Conjugate Priors

We say the family $\lambda_{\mu,k}$ is a *conjugate prior* to p_η if, whenever we take a prior in $\lambda_{\mu,k}$, the posterior is still in the $\lambda_{\mu,k}$ family.

For example,

- The Beta is conjugate to the Binomial.
- The Normal is conjugate to the Normal location family.
- The Gamma is conjugate to the Poisson.

Note

Conjugate priors are not unique! They depend on the carrier you choose. For example, the truncated Gaussian and the Gaussian are both conjugate to the same model!

Contrasts of Bayesian and Frequentist

Frequentist: :(We have to prioritize all θs at once.

Bayesian: :) We can prioritize θ based on a distribution! We can say things have a chance of occurring. It's all computation! The downside is that you actually have to choose a prior, and we will see why that can be hard in a bit.

Question: What is probability really? Mathematically, it is a measure, but in real life, it is usually thought of as a long run frequency, or a proportion from a systematic random sample.

What about questions like "What is the chance that $P = NP$?" This is subjective uncertainty.

Bayesian Law of Large Numbers: Different priors converge to the true value given enough observations.

Question: Why not always be a Bayesian?

Example

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} p$, where p is an unknown density on $\mathbb{R}$, and suppose our estimand is $m = \text{median}(p)$. If n is odd, then $X_{\left(\frac{n+1}{2}\right)}$ is a good estimator. However, to get a Bayes estimator,

1. We need a prior over p (an infinite-dimensional prior!)
2. Compute the posterior :(
3. Compute $\hat{m} = \mathbb{E}[m \mid X]$

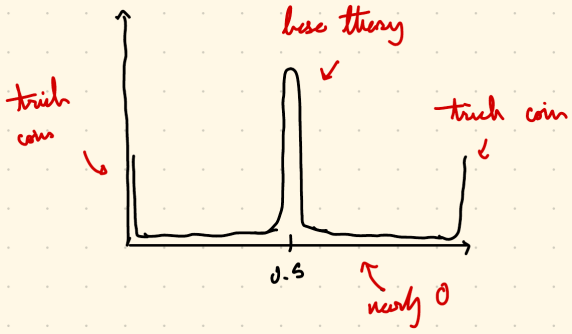
The problem with Bayes is choice paralysis. Frequentist methods are nice because we don't care what p is.

10/13/26 Lecture 14: Objective Bayes

Subjective Priors: These are good if you want your results to reflect your beliefs, but bad if you don't. They are also hard to use if you have high dimensional data, since joint distributions are tough to specify.

Example: Subjective Priors with a Coin

Here is what a subjective prior on the probability that a coin lands heads might look like.



- ✓ Reflects beliefs.
- × Reflects beliefs.

Objective Bayes

Definition 14.1: Flat Prior

The *flat prior* is $\lambda(\theta) \propto_\theta 1$ on Θ.

Note that the flat prior could be improper ($\Lambda(\Theta) = \infty$), but this may not be an issue.

Example: Flat Prior on a Normal Mean

Let θ have the flat prior and $X \mid \theta \sim N(\theta, \sigma^2)$. Then,

$$\lambda(\theta \mid X) \propto_\theta 1 \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\theta)^2}$$

so $\theta \mid X \sim N(X, \sigma^2)$.

Problem: The flat prior is not invariant to reparametrization.

Example

Let $X \sim \text{Binom}(n, \theta)$ with $\theta \in [0, 1]$, and let $\eta = \log\left(\frac{\theta}{1-\theta}\right)$. Then, a flat prior on θ does not give a flat prior on η.

Definition 14.2: Jeffreys' Prior

The *Jeffreys' prior* is

$$\lambda(\theta) \propto |J(\theta)|^{1/2}$$

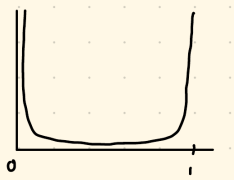
The intuition behind choosing a Jefferey's prior is that if $J(\theta)$ is large, we want to add more density, and if $J(\theta)$ is small, we add less. We are "spreading" out the data in a statistical sense.

Example: Jeffreys' Prior for the Binomial

Let $X \sim \text{Binom}(n, \theta)$ with $\theta \in [0, 1]$. Then,

$$\lambda(\theta) \propto_\theta J(\theta)^{1/2} = \left(\frac{n}{\theta(1-\theta)}\right)^{1/2} \propto_\theta \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right)$$

which looks like



In the θ parametrization, the edges have more information.

Question: How is this objective?
Answer: It's not really, but when we have a default rule to choose the prior, we call it an objective prior.

Intersubjective Agreement

Example

Let $X \sim \text{Bin}(10{,}000, \theta)$ and suppose we observe $X = 3000$, so $\frac{X}{n} = 0.3$. Then,

$$\text{SD}_\theta\left(\frac{X}{n}\right) = \sqrt{\frac{\theta(1-\theta)}{n}} \leq 0.005$$

so the data "swamps" our prior, and we find that $\text{Lik}(\theta; X) \approx 0$ outside of $[0.29, 0.31]$. Then, we can say that

$$\lambda(\theta \mid X) \tilde{\propto}_\theta \text{Lik}(\theta; X) \tilde{\propto}_\theta \exp\left\{-\frac{J(0.3)}{2}(\theta - 0.3)^2\right\}$$

by a Taylor expansion, and the right hand side is proportional to the density of $N(0.3, J(0.3)^{-1})$. Thus,

$$\lambda(\theta \mid X) \approx N(0.3, J(0.3)^{-1}) = N\left(0.3, \frac{0.21}{n}\right) = N(0.3, 2.1 \cdot 10^{-5})$$

Idea: The data pins θ down to a little interval. This is "intersubjective" in the sense that it is the posterior for almost every prior that gives any mass near 0.3 (the data swamps all priors equally).

Example

Let $X \mid \mu \sim N(\mu, I_d)$ with $\mu \in \mathbb{R}^d$. The Fisher information is constant, so the Jeffreys' prior is flat on $\mathbb{R}^d$. Then,

$$\lambda(\mu \mid X) \propto_\mu \exp\left\{-\frac{\|X-\mu\|^2}{2}\right\} \sim N(X, I_d)$$

so the Bayes estimate is $\mathbb{E}[\mu \mid X] = X$.

What about $\rho^2 = \|\mu\|^2$? If $\mu \mid X \sim N_d(X, I_d)$, then

$$\mathbb{E}[\mu_i^2 \mid X] = \mathbb{E}[\mu_i \mid X]^2 + \mathrm{Var}(\mu_i \mid X)$$

so

$$\mathbb{E}[\|\mu\|^2 \mid X] = \|X\|^2 + d$$

Recall that $\delta_{\mathrm{UMVU}}(X) = \|X\|^2 - d$. Notice that $\delta_{\mathrm{UMVU}}(X)$ has the same variance as the Bayes estimator but less bias, so the Bayes estimator is not great! That is, the Jeffreys' prior on μ is not great here.

Indeed, the effective prior induced on ρ^2 is

$$\lambda(\rho^2) \propto_{\rho^2} \rho^{d-2}$$

which makes sense, as it puts more weight on high norms.

Hierarchical and Empirical Bayes

Example: Same-Side Bias

How do we estimate the same-side bias for $m = 48$ coin flippers? Let $\alpha, \beta \sim \lambda$, and

$$\theta_i \mid \alpha, \beta \overset{\text{iid}}{\sim} \mathrm{Beta}(\alpha, \beta) \quad \text{and} \quad X_i \mid \alpha, \beta, \theta \overset{\text{ind}}{\sim} \mathrm{Binom}(n_i, \theta_i)$$

We can find empirically justified α, β using our data from the X_i. If we fix α and β, then

$$\mathbb{E}[\theta_i \mid X, \alpha, \beta] = \frac{X_i + \alpha}{n_i + \alpha + \beta}$$

However, averaging over α and β,

$$\mathbb{E}[\theta_i \mid X] = \iint \frac{X_i + \alpha}{n_i + \alpha + \beta} \lambda(\alpha, \beta \mid X) d\alpha d\beta$$

Here, α and β are *hyper-parameters* that give a "prior" on θ.

This is useful if we don't know much about the θ_i (does matter (much) prior we use).

10/15/26 Lecture 15: Hierarchical Models and Markov Chain Monte Carlo

Example: Batting Averages

Suppose we want to predict a batter's "true" batting average from n_i at-bats. Suppose $X_i \mid \theta_i \overset{\text{ind}}{\sim} \text{Binom}(n_i, \theta_i)$. We want to shrink θ_i to more reasonable values based on historical data, so say that typically,

$$\theta_i \mid \alpha, \beta \overset{\text{iid}}{\sim} \text{Beta}(\alpha, \beta)$$

Now, if we don't know α and β, we can assume $\alpha, \beta \sim \lambda$ and learn α and β from prior data. For a particular α, β, the posterior mean is $\frac{X_i+\alpha}{n_i+\alpha+\beta}$, so when α, β are not known, we will consider the posterior mean

$$\mathbb{E}\left[\frac{X_i+\alpha}{n_i+\alpha+\beta} \mid X\right]$$

Example: Gaussian Hierarchical Models

Let $\tau^2 \sim \lambda_0$, $\theta_i \mid \tau^2 \overset{\text{iid}}{\sim} N(0, \tau^2)$, and $X_i \mid \theta, \tau^2 \overset{\text{ind}}{\sim} N(\theta_i, 1)$ for $i \le d$.
Then, the posterior mean is

$$\begin{aligned}
\delta_i(X) &= \mathbb{E}[\theta_i \mid X] \\
&= \mathbb{E}\left[\mathbb{E}[\theta_i \mid X, \tau^2] \mid X\right] \\
&= \mathbb{E}\left[\frac{\tau^2}{1+\tau^2} X_i \mid X\right] \\
&= \mathbb{E}\left[\frac{\tau^2}{1+\tau^2} \mid X\right] X_i
\end{aligned}$$

where $\mathbb{E}\left[\frac{\tau^2}{1+\tau^2} \mid X\right]$ is the *shrinkage factor*.

Consider $X_i \overset{\text{iid}}{\sim} N(0, 1+\tau^2)$. Then,

$$\frac{1}{d}\|X\|^2 \sim \frac{1+\tau^2}{d}\chi^2_d \sim \left(1+\tau^2, \frac{2(1+\tau^2)^2}{d}\right)$$

where the notation (a, b) means a distribution with mean a and variance b.

Let $\zeta(\tau^2) = \frac{1}{1+\tau^2}$, so $\delta_i(X) = (1 - \mathbb{E}[\zeta \mid X]) \cdot X_i$. We can see $X_i \sim N(0, \zeta^{-1})$. Now, say that

$$\zeta \sim \frac{1}{s^2}\chi^2_k = \text{Gamma}\left(\frac{k}{2}, \frac{2}{s^2}\right) \propto \frac{(s^2)^{k/2}}{\Gamma(k/2)}\zeta^{k/2-1}e^{-s^2\zeta/2}$$

and

$$X \mid \zeta \sim N_d\left(0, \frac{1}{\zeta} I_d\right) \propto_\zeta \zeta^{d/2} e^{-\zeta\|X\|^2/2}$$

Multiplying the prior times the likelihood to get the posterior,

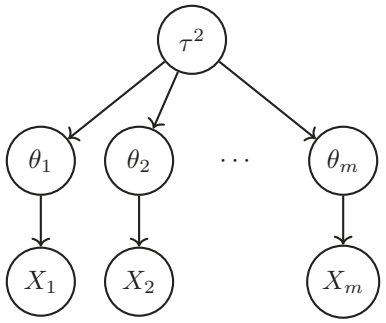
$$\begin{aligned}\lambda(\zeta \mid X) &\propto_\zeta \zeta^{(k+d)/2-1} e^{-(s^2+\|X\|^2)\cdot\zeta/2} \\ &\propto_\zeta \text{Gamma}\left(\frac{k+d}{2}, \frac{2}{s^2+\|X\|^2}\right) \\ &= \frac{1}{s^2+\|X\|^2}\chi^2_{k+d}\end{aligned}$$

This is like adding pseudo-data: we add more degrees of freedom and scale the variance.

The prior expectation is $\frac{k}{s^2}$, and the posterior expectation is $\frac{k+d}{s^2+\|X\|^2}$, so

$$\delta(X) = \left(1 - \frac{k+d}{s^2+\|X\|^2}\right) X_i$$

Graphical Idea:



Heuristically,

1. Generate τ^2.
2. Generate $\theta_1, \ldots, \theta_m$ (independently).
3. Generate $X_1, \ldots, X_m$ (independently).

In general, suppose we have a DAG (V, E) with $V = \{1, \ldots, |V|\}$. Then, if each vertex is a random variable $Z_1, \ldots, Z_{|V|}$, then

$$p(z_1, \ldots, z_{|V|}) = \prod_{i=1}^{|V|} p(z_i \mid z_{\text{Pa}(i)})$$

where $Pa(i) = \{j : j \to i\}$ is the set of parents of i. Then, in our case,

$$p(\tau^2, \theta_1, \ldots, \theta_m, X_1, \ldots, X_m) = p(\tau^2) \prod_{i=1}^{m} p(\theta_i \mid \tau^2) p(x_i \mid \theta_i)$$

Markov Chain Monte Carlo

Recall that

$$\lambda(\theta \mid x) = \frac{p_\theta(x)\lambda(\theta)}{\int_\Theta p_\zeta(x)\lambda(\zeta)d\zeta}$$

The numerator is nice, but the denominator is often intractable.

Computational Strategy: Set up a Markov chain with stationary distribution proportional to $p_\theta(x)\lambda(\theta)$, and run it to get approximate samples.

Definition 15.1: Stationary Markov Chains

A *stationary Markov chain* with transition kernel $Q(y \mid x)$ and initial distribution $\pi_0(x)$ is a sequence of random variables $X^{(0)}, X^{(1)}, X^{(2)}, \dots$ where $X^{(0)} \sim \pi_0$ and

$$X^{(t)} \mid X^{(0)}, \dots, X^{(t-1)} \sim Q(\cdot \mid X^{(t-1)})$$

so $Q(y \mid x) = \mathbb{P}(X^{(t)} = y \mid X^{(t-1)} = x)$.

Note: A Markov chain is an example of a graphical model:

$$X^{(0)} \longrightarrow X^{(1)} \longrightarrow \cdots$$

We see that

$$\pi_1(y) = p(X^{(1)} = y) = \int_{\mathcal{X}} Q(y \mid x)\pi_0(x)d\mu(x)$$

where $Q(y \mid x)\pi_0(x)$ is the probability that we start at x and move to y.

Definition 15.2: Stationary Distributions

We say π is a *stationary distribution*, or is *stationary for* Q, if

$$\pi(y) = \int_{\mathcal{X}} Q(y \mid x)\pi(x)d\mu(x)$$

That is, if we start in π, we stay in π.

Definition 15.3: Detailed Balance

We say π has *detailed balance* if

$$\pi(x)Q(y \mid x) = \pi(y)Q(x \mid y)$$

for all x, y. That is, $P(\text{start at } x \text{ and flow to } y) = P(\text{start at } y \text{ and flow to } x)$. If we have detailed balance, we call the chain *reversible*:

$$X^{(0)} \longrightarrow X^{(1)} \overset{d}{=} X^{(0)} \longleftarrow X^{(1)}$$

Lemma 15.4

If π has detailed balance, then π is stationary.

Proof: We see that

$$\int_{\mathcal{X}} Q(y \mid x)\pi(x)d\mu(x) = \int_{\mathcal{X}} \pi(y)Q(x \mid y)d\mu(x) = \pi(y)\int_{\mathcal{X}} Q(x \mid y)d\mu(x) = \pi(y)$$

as $\int_{\mathcal{X}} Q(x \mid y)d\mu(x) = 1$. ■

Note: Markov chains are forgetful, as the distribution of $X^{(t)}$ only depends on $X^{(t-1)}$.

Theorem 15.5 Markov Chain Convergence Theorem

Suppose a Markov chain with stationary distribution π is

1. **Irreducible:** For all x, y, there exists n such that $p(X^{(n)} = y \mid X^{(0)} = x) > 0$ (for continuous state spaces, we consider neighborhoods of y).
2. **Aperiodic:** For all x, $\gcd\{n > 0 : p(X^{(n)} = x \mid X^{(0)} = x) > 0\} = 1$.

Then, $\text{Dist}(X^{(t)}) \xrightarrow{t\to\infty} \pi$ (in total variation distance) regardless of π_0.

Letting our stationary distribution be π and running the Markov chain, we can approximate properties of its distribution. This method is called *Markov Chain Monte Carlo* (MCMC).

Gibbs Sampler

Suppose we have a parameter vector $(\theta_1, \dots, \theta_d)$.

Gibbs Sampler Algorithm:

Initialize $\theta = \theta^{(0)} \sim \pi_0$
For $t = 1, \dots, T$:
  For $j = 1, \dots, d$:
    Sample $\theta_j \sim \lambda(\theta_j \mid \theta_{-j}, X)$ $(*)$
  Record $\theta^{(t)} = \theta$

Here, θ_{-j} denotes all coordinates but θ_j, and $(*)$ updates the coordinates in the order $1 \to 2 \to \cdots \to d$. Instead of $(*)$, we could alternatively

a) Update 1 random coordinate.

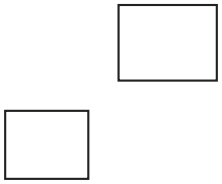
b) Update all coordinates in a random order.

10/20/26 Lecture 16: Gibbs Sampling in Practice and Empirical Bayes

For MCMC, we choose Q so $\pi(\theta) \propto_\theta \lambda(\theta) p_\theta(x)$, then $\text{Dist}(\theta^{(t)}) \to \lambda(\theta \mid x)$.

Recall the Gibbs sampler, where we build $\theta^{(t+1)}$ by looping over each θ_j, sampling from $\lambda(\theta_j \mid \theta_{-j}, X)$.

Note that the Gibbs sampler is not necessarily irreducible. Consider the uniform distribution on the following set:



Then, changing only one coordinate at the time can never take us from one square to another, which implies that starting in one square will force us to stay there.

Claim: For the Gibbs sampler, if $\theta^{(t)} \sim \lambda(\theta \mid X)$, then $\theta^{(t+1)} \sim \lambda(\theta \mid X)$.

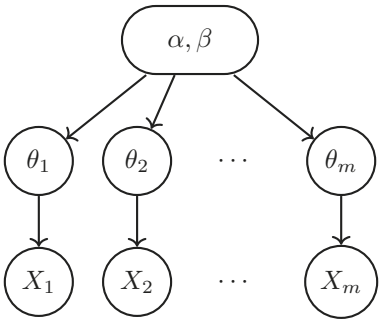
Proof: Sampling θ_j from $\lambda(\theta_j \mid \theta_{-j}, X)$ keeps us in the same distribution, so doing it d times, we are fine. ■

If θ, X follow a DAG model, we can write

$$p(\theta, X) = \prod_{i=1}^{d} p(\theta_i \mid \text{Pa}(\theta_i)) \prod_{i=1}^{n} p(X_i \mid \text{Pa}(X_i))$$

For instance, for the hierarchical model,

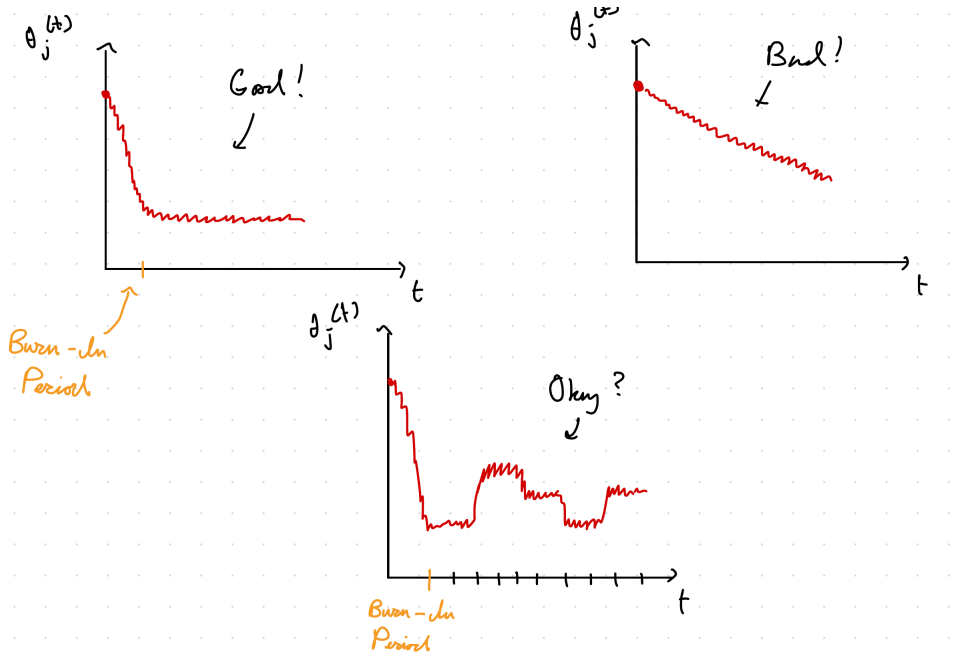
$$\begin{aligned} p(\alpha, \beta, \theta_1, \ldots, \theta_m, X_1, \ldots, X_m) &= p(\alpha, \beta) \prod_{i=1}^{m} p(\theta_i \mid \alpha, \beta) \prod_{i=1}^{m} p(x_i \mid \theta_i) \\ &\propto_{\theta_i} p(\theta_i \mid \alpha, \beta) p(x_i \mid \theta_i) \end{aligned}$$



Gibbs Sampler in Practice

Pro: Given enough draws, we get the distribution.
Con: This may be a very long time.

Plots of Gibbs Samplers



Idea: We discard the entries in the *burn-in period* and *thin* the data by using only every couple samples.

Using Samples: Plot histograms for distribution, take mean for posterior mean. These give us an "empirical estimate" of the posterior.

Empirical Bayes

It's like Hierarchical Bayes, but we learn our hyperpriors:

$$\begin{aligned}\alpha, \beta &\sim \lambda \\ \theta_i \mid \alpha, \beta &\overset{\text{iid}}{\sim} \text{Beta}(\alpha, \beta) \\ X_i \mid \theta_i, \alpha, \beta &\overset{\text{ind}}{\sim} \text{Binom}(n_i, \theta_i)\end{aligned}$$

Learn α and β and hope λ is irrelevant. Thus, we don't really need λ.

Recall our estimator was, under the hierarchical Bayes framework,

$$\delta_i^{\text{HB}}(X) = \mathbb{E}\left[\frac{X_i + \alpha}{n_i + \alpha + \beta} \mid X\right]$$

where α, β are sampled from the posterior distribution $\lambda(\alpha, \beta \mid x)$. For empirical Bayes, we will get

$$\delta_i^{\text{EB}}(X) = \frac{X_i + \hat{\alpha}}{n_i + \hat{\alpha} + \hat{\beta}}$$

where α, β are unknown and we estimate them.

Gaussian Sequence Model

Let $X \sim N_d(\theta, I_d) \iff X_i \overset{\text{ind}}{\sim} N(\theta_i, 1)$.

Goal: Estimate $\theta \in \mathbb{R}^d$ via $\delta(X)$, and we want low $\text{MSE}(\theta; \delta) = \mathbb{E}_\theta \|\theta - \delta(X)\|^2$.

The reason this is good to study is the CLT.

Example: CLT

If $X_1, \dots, X_n \overset{\text{iid}}{\sim} (\theta, \sigma^2 I_d)$, then

$$Z = \frac{1}{\sigma\sqrt{n}} \sum_i X_i \approx N\left(\frac{\sqrt{n}\theta}{\sigma}, I_d\right)$$

Now, the UMVU of the Gaussian sequence model is X. It's also the objective Bayes estimator and the MLE, which is nice.

Linear Shrinkage Estimator:

$$\delta_\zeta(X) = (1 - \zeta)X$$

We saw this to be the Bayes estimator with $\theta_i \overset{\text{ind}}{\sim} N(0, \tau^2)$, where $\zeta = \frac{1}{1+\tau^2}$.

Hierarchical Bayes: If $\tau^2 \sim \lambda_0$, $\theta_i \mid \tau^2 \overset{\text{iid}}{\sim} N(0, \tau^2)$, we would get δ_ζ with $\zeta = \mathbb{E}\left[\frac{1}{1+\tau^2} \mid X\right]$.

Empirical Bayes: Estimate τ^2 or estimate ζ directly, so we get $\delta_{\hat{\zeta}}$ or δ_ζ with $\zeta = \frac{1}{1+\hat{\tau}^2}$. Since $X_i \mid \tau^2 \overset{\text{iid}}{\sim} N(0, 1 + \tau^2)$,

$$\implies \|X\|^2 \sim (1 + \tau^2)\chi^2_d \implies \mathbb{E}[\|X\|^2] = (1 + \tau^2)d$$

so estimate $\hat{\tau}^2 = \frac{\mathbb{E}[\|X\|^2]}{d} - 1$, so $\hat{\zeta} = \frac{d}{\mathbb{E}[\|X\|^2]}$.

Note: $\hat{\zeta}$ is positively biased by Jensen's, but it's still the MLE estimate.

10/22/26 Lecture 17: The James-Stein Estimator and Stein's Lemma

Proposition 17.1

If $Y \sim \chi^2_d$, then $\mathbb{E}Y = d$ and $\mathbb{E}[1/Y] = 1/(d-2)$ for $d \geq 3$.

Proof:

$$\begin{aligned}
\mathbb{E}[1/Y] &= \int_0^\infty \frac{1}{y} \cdot \frac{1}{2^{d/2}\Gamma(d/2)} y^{d/2-1} e^{-y/2} dy \\
&= \int_0^\infty \frac{1}{2^{d/2}\Gamma(d/2)} y^{\left(\frac{d-2}{2}\right)-1} e^{-y/2} dy \\
&= \frac{2^{(d-2)/2}\Gamma\left(\frac{d-2}{2}\right)}{2^{d/2}\Gamma(d/2)} \int_0^\infty \frac{1}{2^{(d-2)/2}\Gamma\left(\frac{d-2}{2}\right)} y^{\left(\frac{d-2}{2}\right)-1} e^{-y/2} dy \\
&= \frac{\Gamma\left(\frac{d}{2}-1\right)}{2 \cdot \Gamma(d/2)} \\
&= \frac{1}{2 \cdot \left(\frac{d}{2}-1\right)} = \frac{1}{d-2}
\end{aligned}$$

■

Recall: We had $X \sim N_d(\theta, I_d)$. The obvious estimator $\delta(X) = X$ is pretty good. This is δ_0 for

$$\delta_\zeta(X) = (1-\zeta)X \quad (\zeta \in [0,1])$$

If $\theta_i \overset{\text{iid}}{\sim} N(0, \tau^2)$ (τ^2 known), then $\zeta = \frac{1}{1+\tau^2}$ is Bayes.

If we don't know τ^2, then as $X_i \overset{\text{iid}}{\sim} N(0, 1+\tau^2)$, then $\|X\|^2 \sim (1+\tau^2)\chi^2_d$, so the HB estimate may be $\delta_{\zeta_{\text{HB}}}$ for

$$\zeta_{\text{HB}} = \frac{k+d}{s^2 + \|X\|^2}$$

If we don't want HB, then we can use EB to get

$$\hat{\zeta}_{\text{MLE}} = \frac{1}{1+\hat{\tau}^2_{\text{MLE}}} = \frac{d}{\|X\|^2}$$

Finally, if we want $\hat{\zeta}_{\text{UMVUE}}$, since

$$\mathbb{E}\left[\frac{d}{\|X\|^2}\right] = \frac{d}{1+\tau^2}\mathbb{E}[1/Y] = \frac{d}{d-2} \cdot \frac{1}{1+\tau^2} \quad \text{where } Y = \frac{\|X\|^2}{1+\tau^2}$$

so we see $\mathbb{E}\left[\frac{d-2}{\|X\|^2}\right] = \frac{1}{1+\tau^2}$, so

$$\hat{\zeta}_{\text{UMVUE}} = \frac{d-2}{\|X\|^2}$$

Choosing $\hat{\zeta}_{\text{UMVUE}}$, we get what is called the *James-Stein estimator*

$$\delta_{\text{JS}}(X) = \left(1 - \frac{d-2}{\|X\|^2}\right) X$$

Incredibly, δ_{JS} dominates δ_0 for all θ! Thus, X is inadmissible! Thus, adaptive shrinkage is fundamentally strong.

Let's try to prove this. First,

$$\begin{aligned} R(\theta, \delta_\zeta) &= \|\theta - \mathbb{E}\delta_\zeta(X)\|^2 + \sum_i \text{Var}((1-\zeta)X_i) \\ &= \|\theta - (1-\zeta)\theta\|^2 + \sum_i \text{Var}((1-\zeta)X_i) \\ &= \zeta^2\|\theta\|^2 + d(1-\zeta)^2 \end{aligned}$$

Taking derivatives,

$$\begin{aligned} 0 = \frac{d}{d\zeta}R &= 2\zeta\|\theta\|^2 - 2(1-\zeta)d \\ \zeta^*(\theta) &= \frac{d}{d+\|\theta\|^2} = \frac{1}{1+\|\theta\|^2/d} \end{aligned}$$

where $\|\theta\|^2/d = \frac{1}{d}\sum_i \theta_i^2$. This is a function of θ, so we can't get a unanimous ζ^*. However, let's see if we can get a good value. For example, if we set

$$\begin{aligned} \frac{d-2}{\|X\|^2} &= \frac{d}{d+\|\theta\|^2} \\ (d-2)(d+\|\theta\|^2) &= d\|X\|^2 \\ \|\theta\|^2 &= \frac{d(\|X\|^2 - d + 2)}{d-2} \end{aligned}$$

so it induces an estimate on $\|\theta\|^2$.

Theorem 17.2: Stein's Lemma, Univariate

Suppose $X \sim N(\theta, \sigma^2)$ and $h : \mathbb{R} \to \mathbb{R}$ such that h is differentiable and $\mathbb{E}\left|\dot{h}(X)\right| < \infty$. Then,

$$\text{Cov}(X, h(X)) = \mathbb{E}[(X-\theta)h(X)] = \sigma^2\mathbb{E}[\dot{h}(X)]$$

Proof: We can assume WLOG $h(0) = 0$, since we can shift and nothing changes.

We start with when $\theta = 0$ and $\sigma^2 = 1$. Then,

$$\mathbb{E}[Xh(X)] = \int_0^\infty xh(x)\phi(x)dx + \int_{-\infty}^0 xh(x)\phi(x)dx$$

Then,

$$\begin{aligned} \int_0^\infty xh(x)\phi(x)dx &= \int_0^\infty x\left[\int_0^x \dot{h}(y)dy\right]\phi(x)dx \\ &= \int_0^\infty \int_0^\infty 1\{y < x\}x\dot{h}(y)\phi(x)dxdy \\ &= \int_0^\infty \dot{h}(y)\int_y^\infty x\phi(x)dxdy \\ &= \int_0^\infty \dot{h}(y)\phi(y)dy \end{aligned}$$

since $\frac{d}{dx}\phi(x) = -x\phi(x)$, so $\int_y^\infty x\phi(x)dx = [-\phi(x)]_y^\infty = \phi(y)$. So,

$$\mathbb{E}[Xh(X)] = \mathbb{E}[\dot{h}(X)]$$

Now, for $X = \theta + \sigma Z$ for standard Gaussian Z,

$$\begin{aligned}\mathbb{E}[(X-\theta)h(X)] = \mathbb{E}[\sigma Z h(\theta+\sigma Z)] &= \sigma\mathbb{E}[Zh(\theta+\sigma Z)] \\ &= \sigma\mathbb{E}[\sigma\dot{h}(\theta+\sigma Z)] \\ &= \sigma^2\mathbb{E}[\dot{h}(X)]\end{aligned}$$

■

Definition 17.3: Jacobians

If $h : \mathbb{R}^d \to \mathbb{R}^d$, let $Dh : \mathbb{R}^d \to \mathbb{R}^{d^2}$ be its *Jacobian*. That is, the matrix such that

$$(Dh)_{i,j} = \frac{\partial h_i}{\partial x_j}$$

Definition 17.4: Frobenius Norm

If $A \in \mathbb{R}^{d^2}$, let $\|A\|_F^2 = \sum_{i,j} A_{i,j}^2$ be it's *Frobenius norm*.

Theorem 17.5: Stein's Lemma, Multivariate

Let $h : \mathbb{R}^d \to \mathbb{R}^d$, $X \sim N_d(\theta, \sigma^2 I_d)$, $\theta \in \mathbb{R}^d$, and $\mathbb{E}\|Dh\|_F < \infty$. Then,

$$\mathbb{E}[(X-\theta)'h(X)] = \sigma^2\mathbb{E}[\text{tr}(Dh(X))] = \sigma^2\sum_{i=1}^d \mathbb{E}\left[\frac{\partial h_i}{\partial x_i}(X)\right]$$

Proof:

$$\begin{aligned}\mathbb{E}[(X_i-\theta_i)h_i(X)] = \mathbb{E}[\mathbb{E}[(X_i-\theta_i)h_i(X) \mid X_{-i}]] &= \mathbb{E}\left[\mathbb{E}\left[\sigma^2\frac{\partial h_i}{\partial x_i}(X) \mid X_{-i}\right]\right] \\ &= \mathbb{E}\left[\sigma^2\frac{\partial h_i}{\partial x_i}(X)\right]\end{aligned}$$

Then, we get our result after summing over the d terms.

■

Stein's Unbiased Risk Estimator (SURE)

Let $h(X) = X - \delta(X)$. Then,

$$\begin{aligned} R(\theta;\delta) = \mathbb{E}\|\theta - \delta(X)\|^2 &= \mathbb{E}\|X - \theta - h(X)\|^2 \\ &= \mathbb{E}\|X-\theta\|^2 + \mathbb{E}\|h(X)\|^2 - 2\mathbb{E}[(X-\theta)'h(X)] \\ &= d\sigma^2 + \mathbb{E}\|h(X)\|^2 - 2\sigma^2\mathbb{E}_\theta[\operatorname{tr}(Dh(X))] \end{aligned}$$

Then, take

$$\begin{aligned} \hat{R}(X) &= \sigma^2 d + \|X - \delta(X)\|^2 - 2\sigma^2 \operatorname{tr}(Dh(X)) \\ &= \sigma^2 d + \|X-\delta(X)\|^2 - 2\sigma^2 \sum_{i=1}^d \left(1 - \frac{\partial}{\partial x_i}\delta_i(X)\right) \end{aligned}$$

Then, $\hat{R}(X)$ is an unbiased estimate of the MSE, so we can find δ minimizing the expectation of this. We can also use it to graph the MSE by taking its expectation.

Example: Risk of James-Stein

Assume $X \sim N_d(\theta, I_d)$. Recall $\delta_{\text{JS}}(X) = \left(1 - \frac{d-2}{\|X\|^2}\right)X$, so

$$h(X) = \frac{d-2}{\|X\|^2}X \implies \|h(X)\|^2 = \frac{(d-2)^2}{\|X\|^2}$$

Then,

$$\begin{aligned} \frac{\partial h_i}{\partial x_i}(X) = \frac{\partial}{\partial x_i}\frac{(d-2)x_i}{\|x\|^2} &= \frac{\partial}{\partial x_i}\frac{(d-2)x_i}{\sum x_i^2} \\ &= (d-2)\frac{\|X\|^2 - 2X_i^2}{(\|X\|^2)^2} \end{aligned}$$

Then,

$$\sum_{i=1}^d \frac{\partial h_i}{\partial x_i}(X) = \frac{d-2}{\|X\|^4}\left(d\|X\|^2 - 2\|X\|^2\right) = \frac{(d-2)^2}{\|X\|^2}$$

So

$$\begin{aligned} \hat{R}(X) &= d + \frac{(d-2)^2}{\|X\|^2} - 2\frac{(d-2)^2}{\|X\|^2} \\ &= d - \frac{(d-2)^2}{\|X\|^2} \end{aligned}$$

Then,

$$R(\theta, \delta_{\text{JS}}) = \mathbb{E}[\hat{R}(X)] = d - (d-2)^2\mathbb{E}\left[\frac{1}{\|X\|^2}\right] < d = R(\theta, X)$$

so δ_{JS} dominates δ_0. In the $\theta = 0$ case,

$$R(0, \delta_{\text{JS}}) = d - \frac{(d-2)^2}{d-2} = 2$$

10/27/26 Lecture 18: Beyond James-Stein and Minimax Estimation

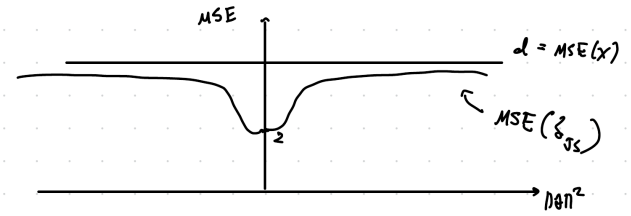
Recall: For $X \sim N_d(\theta, I_d)$, for the shrinkage estimator $(1-\zeta)X$, we found

$$\hat{\zeta}_{\text{UMVU}} = \frac{d-2}{\|X\|^2} \rightsquigarrow \delta_{\text{JS}} = \left(1 - \frac{d-2}{\|X\|^2}\right) X$$

for $\zeta = \frac{1}{1+\tau^2}$, but even without a prior on τ, δ_{JS} still is better than X. We also had

$$\text{MSE}_\theta(\delta_{\text{JS}}) = d - \mathbb{E}_\theta\left[\frac{(d-2)^2}{\|X\|^2}\right]$$

and $\text{MSE}_0(\delta_{\text{JS}}) = 2$. Plotting,



Now, note that $\frac{d-2}{\|X\|^2}$ can be ≥ 1, so an even better estimator is

$$\delta_{\text{JS}^+}(X) = \left(1 - \frac{d-2}{\|X\|^2}\right)_+ X$$

Another thing you can do is consider

$$\delta_{\text{JS},2}(X) = \left(1 - \frac{d-3}{\|X - \overline{X}1_d\|^2}\right)(X - \overline{X}1_d) + \overline{X}1_d$$

for $d \geq 4$.

For $d \geq 4$, $\delta_{\text{JS},2}(X)$ strictly dominates δ_0.

Idea of derivation: Change of basis so that $\overline{X}1_d$ is a basis vector, and apply JS on the $(d-1)$-dimensional subspace orthogonal to it, using the projection onto the subspace.

Minimax Risk

The idea is to find

$$\underset{\delta}{\text{minimize}} \left(\sup_\theta R(\theta, \delta)\right)$$

Definition 18.1: Minimax Risk

The *minimax risk* is $r^* = \inf_\delta \sup_\theta R(\theta, \delta)$. An estimator δ^* is *minimax* if $\sup_\theta R(\theta, \delta^*) = r^*$.

In a sense, this is a very pessimistic estimator, but it has some uses. For example, in game theory, if someone was picking a purposely difficult-to-estimate θ, δ^* is good. It can also be used to bound how hard estimation

of a problem is.

This is also connected to Bayes, as δ^* is okay for all priors. That is, if there is some prior that you don't know, then δ^* is good.

Note: For any proper prior Λ, the Bayes risk is

$$\begin{aligned} r_\Lambda &= \inf_\delta \int R(\theta, \delta) d\Lambda(\theta) \\ &\leq \inf_\delta \sup_\theta R(\theta, \delta) \\ &= r^* \end{aligned}$$

Moreover, if δ_Λ is Bayes, then $r_\Lambda = \int R(\theta, \delta_\Lambda) d\Lambda(\theta)$, so the Bayes risk of any prior lower bounds r^*.

Definition 18.2: Least Favorable Priors

A *least favorable prior* Λ^* gives the best lower bound

$$r_{\Lambda^*} = \sup_\Lambda r_\Lambda$$

so for any Λ and any δ,

$$r_\Lambda \leq r_{\Lambda^*} \leq r^* \leq \sup_\theta R(\theta, \delta)$$

Theorem 18.3: Correspondence between Bayes and Minimax

Let Λ be a prior with Bayes estimator δ_Λ. If $r_\Lambda = \sup_\theta R(\theta; \delta_\Lambda)$, then δ_Λ is minimax, Λ is LF, and $r^* = r_\Lambda$.

Moreover, if δ_Λ is the unique Bayes estimator up to $\overset{\text{a.s.}}{=}$, it is also the unique minimax estimator.

Proof: The first statement is obvious from above:

$$\sup_\theta R(\theta, \delta_\Lambda) = r_\Lambda \leq r^* \leq \sup_\theta R(\theta, \delta_\Lambda)$$

For any other δ,

$$\begin{aligned} \sup_\theta R(\theta, \delta) \geq \int R(\theta, \delta) d\Lambda(\theta) &> \int R(\theta, \delta_\Lambda) d\Lambda(\theta) \\ &= r_\Lambda = r^* \end{aligned}$$

where $>$ comes from uniqueness of δ_Λ.

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Note: The condition $r_\Lambda = \sup_\theta R(\theta; \delta_\Lambda)$ implies we have almost everywhere constant risk on the support of Λ.

Checkable Conditions:

1. δ_Λ is a Bayes estimator with constant risk.
2. δ_Λ is Bayes and its risk attains its max on the support of Λ.

Example: $X \sim N(\theta, 1)$, $|\theta| \geq 1$. We want to estimate $g(\theta) = \text{sgn}(\theta)$.

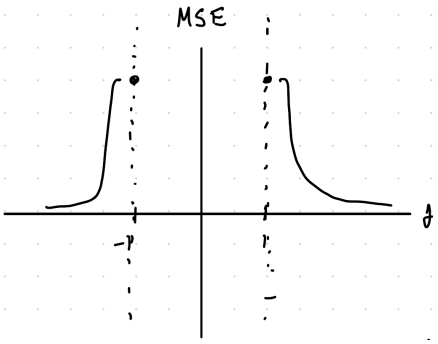
Intuitively, $\Lambda = \text{Unif}(\{-1, 1\})$ is a pretty unfavorable prior, so let's try to find a Bayes estimator for this.

$$\mathbb{P}_\Lambda(\theta = 1 \mid X = x) = \frac{\phi(x-1)}{\phi(x-1) + \phi(x+1)} \quad \text{and} \quad \mathbb{P}_\Lambda(\theta = -1 \mid X = x) = \frac{\phi(x+1)}{\phi(x-1) + \phi(x+1)}$$

so

$$\mathbb{E}_\Lambda(g(\theta) \mid X = x) = \frac{\phi(x-1) - \phi(x+1)}{\phi(x-1) + \phi(x+1)} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \tanh(x)$$

Then, we see



Then, as the risk of δ_Λ is constant on the support of Λ (and maximized there), we see δ_Λ is minimax.

Example: $X \sim \text{Binom}(n, \theta)$, estimate θ using MSE.

Consider $\theta \sim \text{Beta}(\alpha, \beta)$ and $\alpha = \beta$, so $\delta_\alpha(X) = \frac{X+\alpha}{n+2\alpha}$. Now, consider

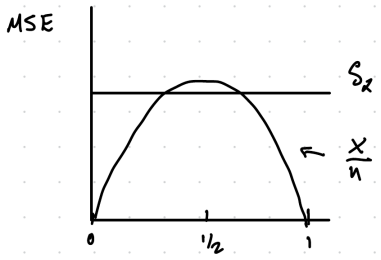
$$\mathbb{E}_\theta \delta_\alpha(X) = \frac{n\theta + \alpha}{n + 2\alpha} \implies \text{Bias}(\theta, \delta_\alpha) = \frac{\alpha(1-2\theta)}{n+2\alpha}$$

$$\text{Var}_\theta(\delta_\alpha(X)) = \frac{n\theta(1-\theta)}{(n+2\alpha)^2}$$

Then

$$\begin{aligned} \text{MSE}(\delta_\alpha) &= \text{Bias}^2 + \text{Var} \\ &= \frac{1}{(n+2\alpha)^2}\left(\alpha^2(1-2\theta)^2 + n\theta(1-\theta)\right) \\ &= (n+2\alpha)^{-2}\left(\alpha^2 + (n-4\alpha^2)\theta(1-\theta)\right) \end{aligned}$$

So we can take $\alpha = \frac{\sqrt{n}}{2}$, which makes this constant in θ. Thus, $\delta_\alpha(X) = \frac{X+\sqrt{n}/2}{n+\sqrt{n}}$ is minimax. However, note this isn't very good by this graph:


MSE
S_x
p
$\frac{X}{n}$
0
1/2
1

10/29/26 Lecture 19: Least Favorable Sequences and Hypothesis Testing

Recall:

$$r^* = \inf_\delta \sup_\theta R(\theta; \delta)$$

may not be attained, but if $\sup_\theta R(\theta; \delta^*) = r^*$, then δ^* is minimax.

Now, recall

$$r_\Lambda = \inf_\delta \int_\Theta R(\theta, \delta) d\Lambda(\theta)$$

and if Λ^* maximizes r_Λ, then Λ^* is least-favorable. Recall that, for all δ, Λ,

$$r_\Lambda \le r_{\Lambda^*} \le r^* \le \sup_\theta R(\theta, \delta)$$

Thus, if we can find a pair Λ, δ_Λ for which

$$r_\Lambda = \sup_\theta R(\theta, \delta_\Lambda)$$

then Λ is least favorable and δ_Λ is minimax.

Thus, δ is minimax if δ is Bayes with constant risk.

Example: Binomial

$X \sim \text{Binom}(n, \theta)$, δ_Λ is minimax for $\Lambda \sim \text{Beta}\left(\frac{\sqrt{n}}{2}, \frac{\sqrt{n}}{2}\right)$.

Example: Gaussian

What about $X \sim N(\theta, 1)$, $\theta \in \mathbb{R}$, and we want to estimate θ with squared error loss?

Idea: Flat prior is good but not proper.

This motivates LF prior sequences.

Definition 19.1: Least Favorable Sequences

A sequence $\Lambda_1, \Lambda_2, \dots$ is *least favorable* if $r_{\Lambda_k} \to \sup_\Lambda r_\Lambda$.

Theorem 19.2

Suppose $\Lambda_1, \Lambda_2, \dots$ is a sequence of priors and δ is an estimator such that

$$\sup_\theta R(\theta, \delta) = \lim_{n \to \infty} r_{\Lambda_n}$$

Then, $\{\Lambda_i\}$ is LF, δ is minimax, and $r^* = \lim_n r_{\Lambda_n}$.

Proof:

$$r_{\Lambda_i} \le \sup_{\Lambda} r_\Lambda \le r^* \le \sup_\theta R(\theta, \delta)$$

and $r_{\Lambda_i} \to \sup_\theta R(\theta, \delta)$.

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Example: $X \sim N_d(\theta, I_d)$, $\delta_0(X) = X$, and MSE $\equiv d$. Now, we see that $\Lambda_k = N_d(0, kI_d)$ converges to the flat prior. Recall that the Bayes estimator is

$$\delta_{\Lambda_k}(X) = (1 - \zeta_k)X, \quad \zeta_k = \frac{1}{1+k}$$

and so

$$\text{MSE}(\theta, \delta_{\Lambda_k}) = \zeta_k^2 \|\theta\|^2 + (1-\zeta_k)^2 d$$

Taking $k \to \infty$, we see

$$r_{\Lambda_k} = \mathbb{E}[\text{MSE}(\theta, \delta_{\Lambda_k})] = \frac{kd}{(1+k)^2} + \frac{k^2 d}{(1+k)^2} \to d$$

so as $\sup_\theta R(\theta, \delta_0) = d = \lim_{k\to\infty} r_{\Lambda_k}$, we see δ_0 is minimax and $(N_d(0, kI_d))_k$ is LF.

Bounding the Minimax Risk

Notice that our theorem gives us a way to bound r^*, as it tells us that for any δ, Λ,

$$\int R(\theta; \delta_\Lambda) d\Lambda(\theta) \le r^* \le \sup_\theta R(\theta; \delta)$$

thus, although minimax esimators are typically hard to find, minimax bounds are often used to characterize how hard a problem is (especially the lower bound).

That is, we will typically propose a practical estimator δ, and find a Λ for which r_Λ is close to $\sup_\theta R(\theta; \delta)$. Then, if we can do that, we can say that δ cannot be improved much in the minimax sense.

One caveat with minimax estimators however is that a problem might be typically easy throughout most of the parameter space, but very difficult in a bizarre case that is rarely every encountered.

Hypothesis Testing

Setup: Data $X \sim P_\theta$, P_θ in model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$. Partition into:

$$\begin{aligned} \text{Null Hyp.} \quad & H_0 : \theta \in \Theta_0 \\ \text{Alternative Hyp.} \quad & H_1 : \theta \in \Theta_1 \end{aligned}$$

Usually: Require hypotheses to be disjoint and exhaustive (so Θ_0, Θ_1 partition Θ). If Θ_1 is unspecified, we can let $\Theta_1 = \Theta \setminus \Theta_0$.

We have two actions:
Reject H_0: Conclude $\theta \in \Theta_1$
Fail to reject H_0: No conclusion

Typically, P_θ has the same support for all θ, so we can never completely rule out $\theta \in \Theta_0$. Thus, we typically never have complete certainty in our actions.

Definition 19.3: Simple and Composite Hypotheses

H_0 or H_1 are called *simple* if $\Theta_0 = \{\theta_0\}$ or $\Theta_1 = \{\theta_1\}$. Otherwise, we call them *composite*.

Simple hypotheses are nice since we just need to consider one distribution; namely P_{θ_0} or P_{θ_1}.

Example: z-test

$X \sim N(\theta, 1)$.

$$H_0 : \theta \leq 0 \quad \text{vs} \quad H_1 : \theta > 0 \quad \text{(comp vs comp)}$$
$$H_0 : \theta = 0 \quad \text{vs} \quad H_1 : \theta \neq 0 \quad \text{(simple vs comp)}$$

Example: Two-Sample

$X_1, \dots, X_n \overset{\text{iid}}{\sim} P$ and $Y_1, \dots, Y_m \overset{\text{iid}}{\sim} Q$.

$$H_0 : P = Q \quad \text{vs} \quad H_1 : P \neq Q$$

To make these decisions, we construct the critical function

$$\phi(X) = \begin{cases} 0 & \text{accept } H_0 \\ \pi \in (0,1) & \text{reject } H_0 \text{ w.p. } \pi \\ 1 & \text{reject } H_0 \end{cases}$$

If a critical function is not randomized, we partition $\mathcal{X}$ into

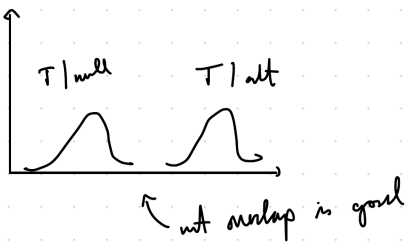
$$\text{rejection region: } R = \{x : \phi(x) = 1\} \qquad \text{acceptance region: } A = \{x : \phi(x) = 0\}$$

A useful way to construct a test is via a *test statistic* $T(X) \in \mathbb{R}$. We say ϕ *rejects for large* $T(X)$ if

$$\phi(x) = \begin{cases} 0 & T(x) < c \\ \gamma \in (0,1) & T(x) = c \\ 1 & T(x) > c \end{cases} \quad \text{(if randomized)}$$

We say c is the *critical threshold*.

Thus, the goal is to choose a good test statistic T to discriminate H_0 from H_1.



Significance Level

Type I Error: H_0 true, but we reject
Type II Error: H_0 false, but we don't reject

Typically we, as statisticians, are required to control Type I, but subject to a constraint on Type I error, we minimize Type II error.

Usual goal: Minimize $\mathbb{P}_{H_1}(\text{Type II})$ subject to $\mathbb{P}_{H_0}(\text{Type I}) \leq \alpha$.

Power function: $\beta_\phi(\theta) = \mathbb{E}_\theta[\phi(X)] = \mathbb{P}_\theta(\text{Reject } H_0)$. Then, the goal is

$$\underset{\phi}{\text{maximize}}\ \beta_\phi(\theta) \text{ for } \theta \in \Theta_1 \quad \text{subject to} \quad \beta_\phi(\theta) \leq \alpha \text{ for } \theta \in \Theta_0$$

Definition 19.4: Level-α Tests

We say ϕ is a *level-α test* if $\sup_{\theta \in \Theta_0} \beta_\phi(\theta) \leq \alpha$.